

Mobile Applications

Fall 2025

Prerequisites

- Modern programming language
- Object oriented
- Statically types
- IDE - IntelliJ/Android Studio or Visual Studio Code



What should you know...

- Basics:
 - Object-oriented programming
 - Classes, methods
 - Exception handling



Bonus

- Functional Programming
- Lambdas
- Higher Order Functions
- Reactive Programming



Options



2007



2008



2010



Native Options



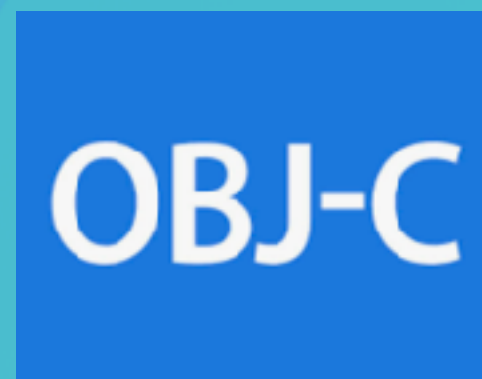
2008



2007



2017



2014

Non-Native Options



2013

Hybrid App

Non-Native Options



Compiled App



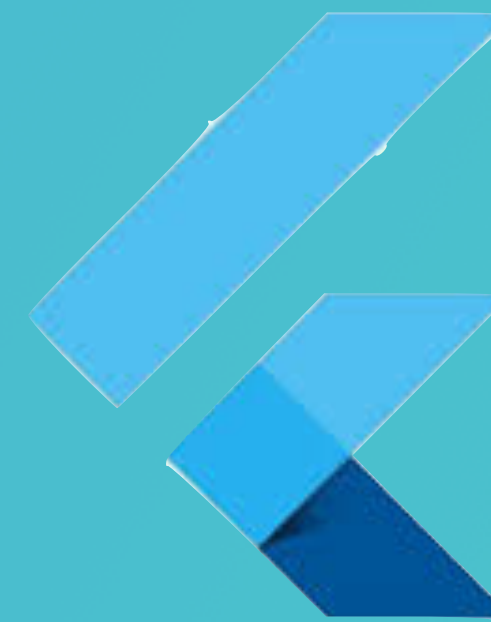
2014

JS



2015

JS



2017



Timeline



2007

2008

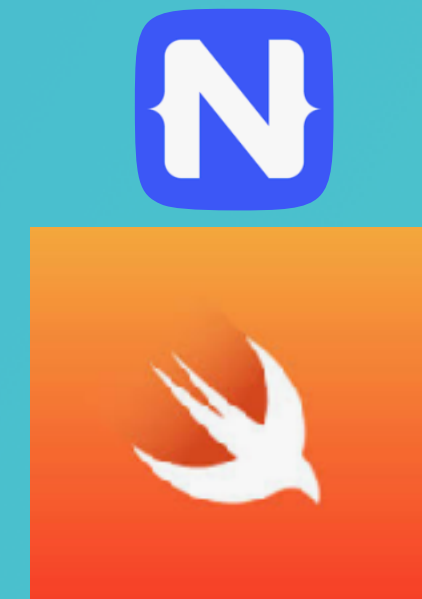


2011



2013

2014



2015



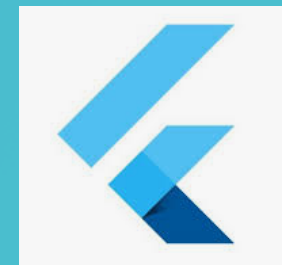
2017



What to learn?



OBJ-C



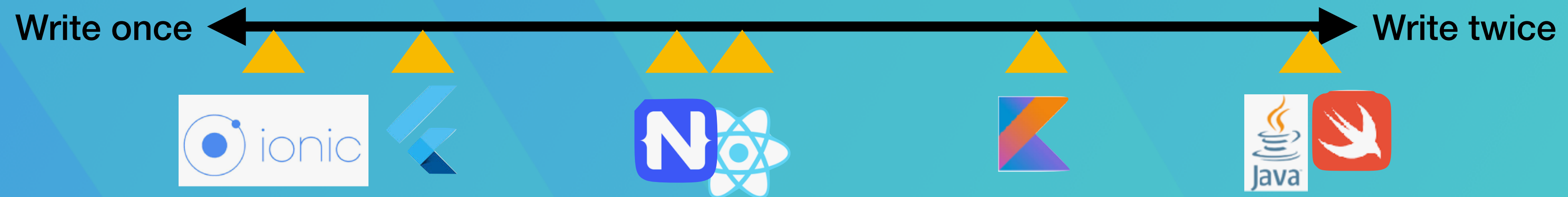
KNOW YOUR RIGHTS

bit.ly/maQuiz2025



Comparison

Write once, use everywhere



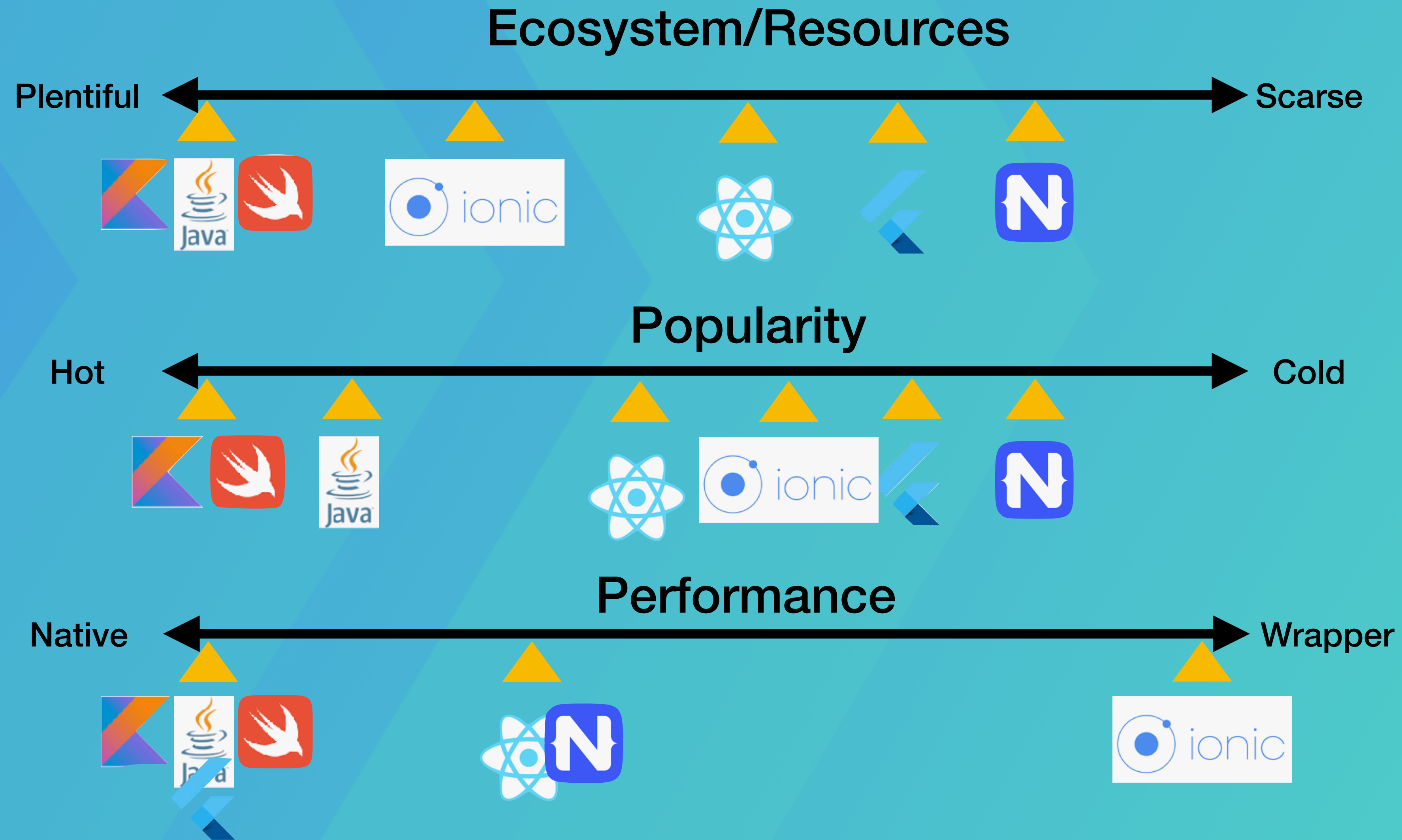
Learn once, write everywhere



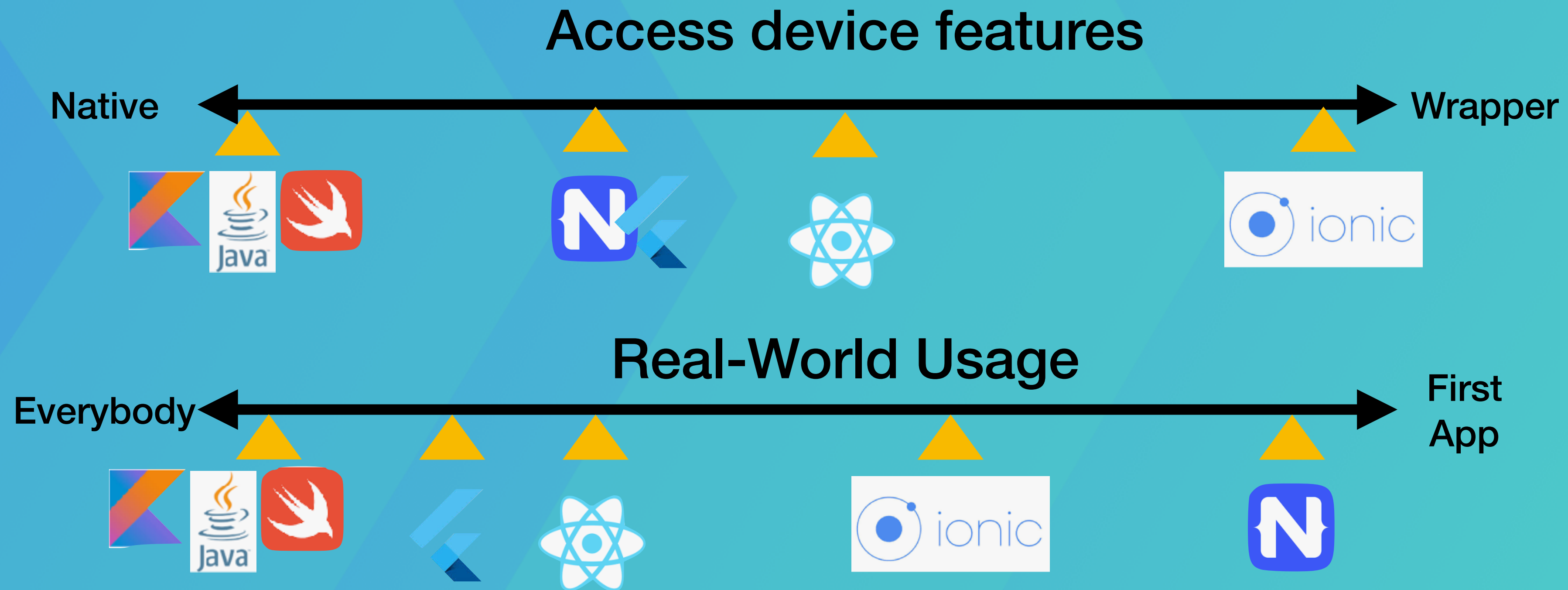
Rich component library



Comparison



Comparison



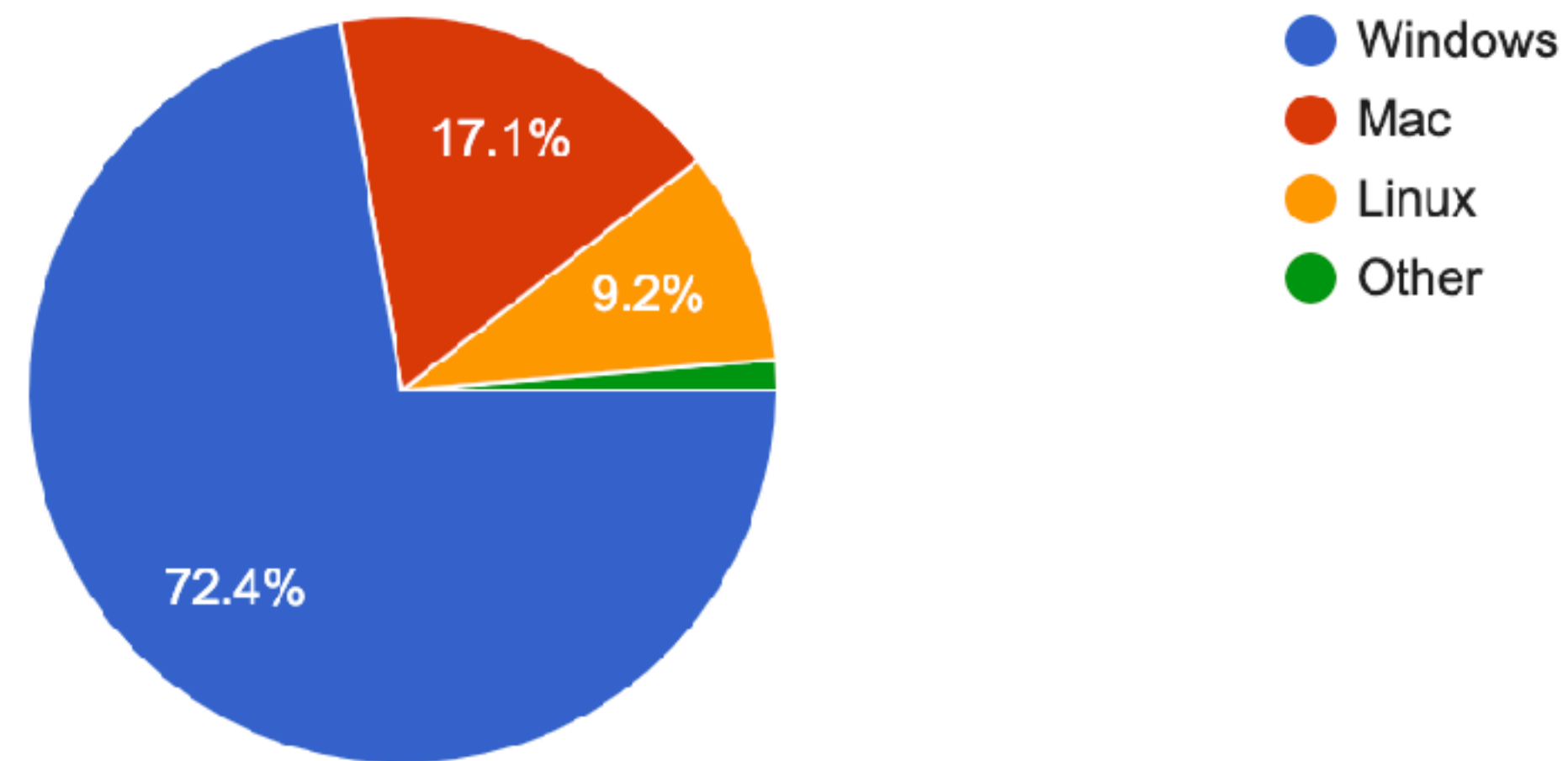
Previous years



Responses from 2018

What operating system is your development machine using?

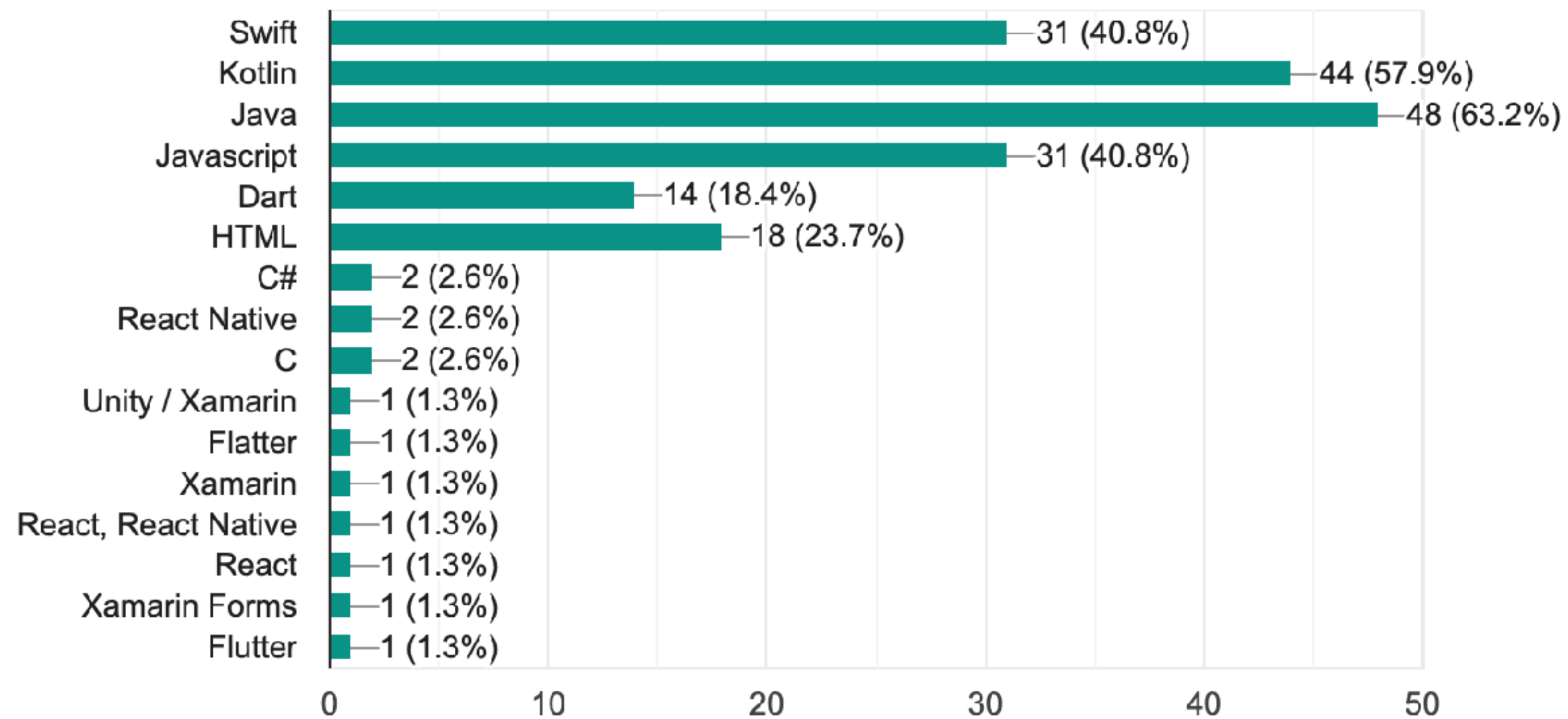
76 responses



Responses from 2018

What language would you like to use/learn?

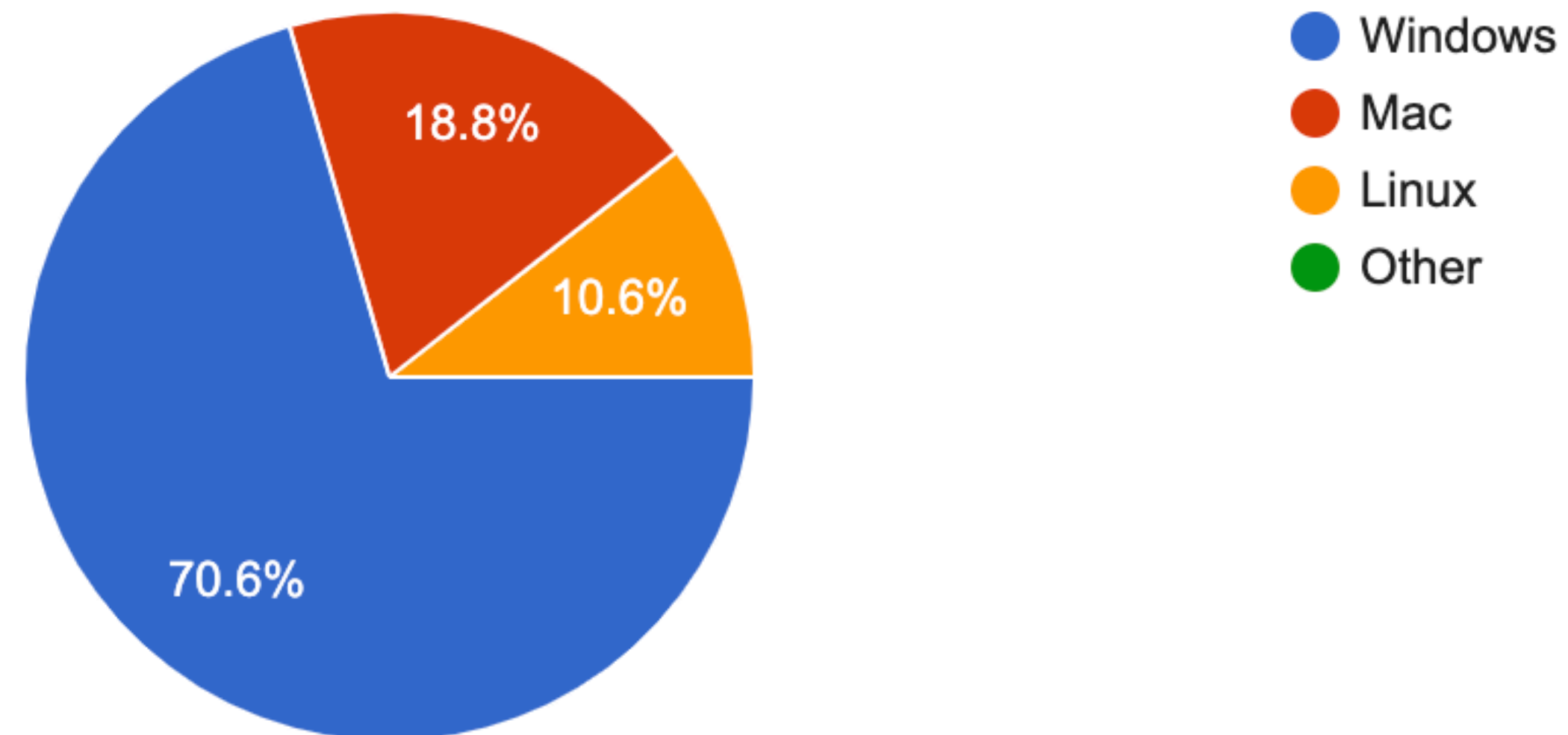
76 responses



Responses from 2019

What operating system is your development machine using?

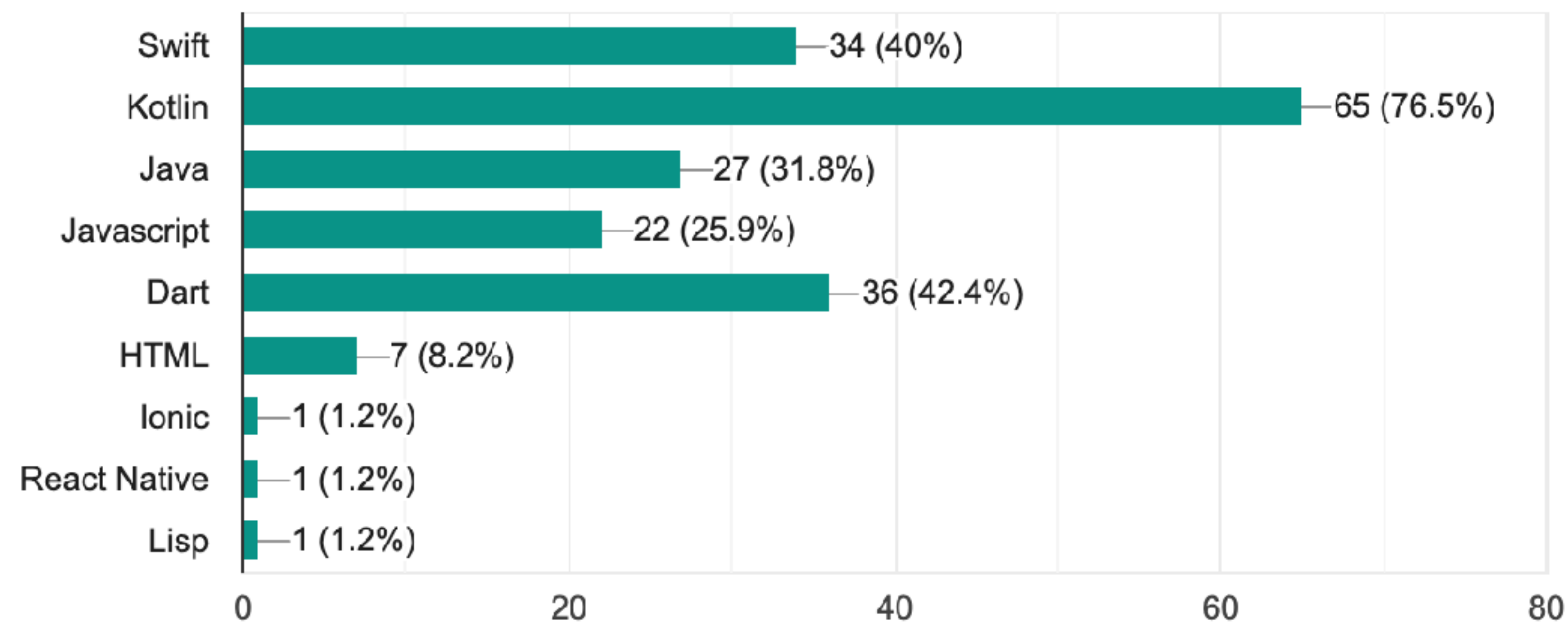
85 responses



Responses from 2019

What language would you like to use/learn?

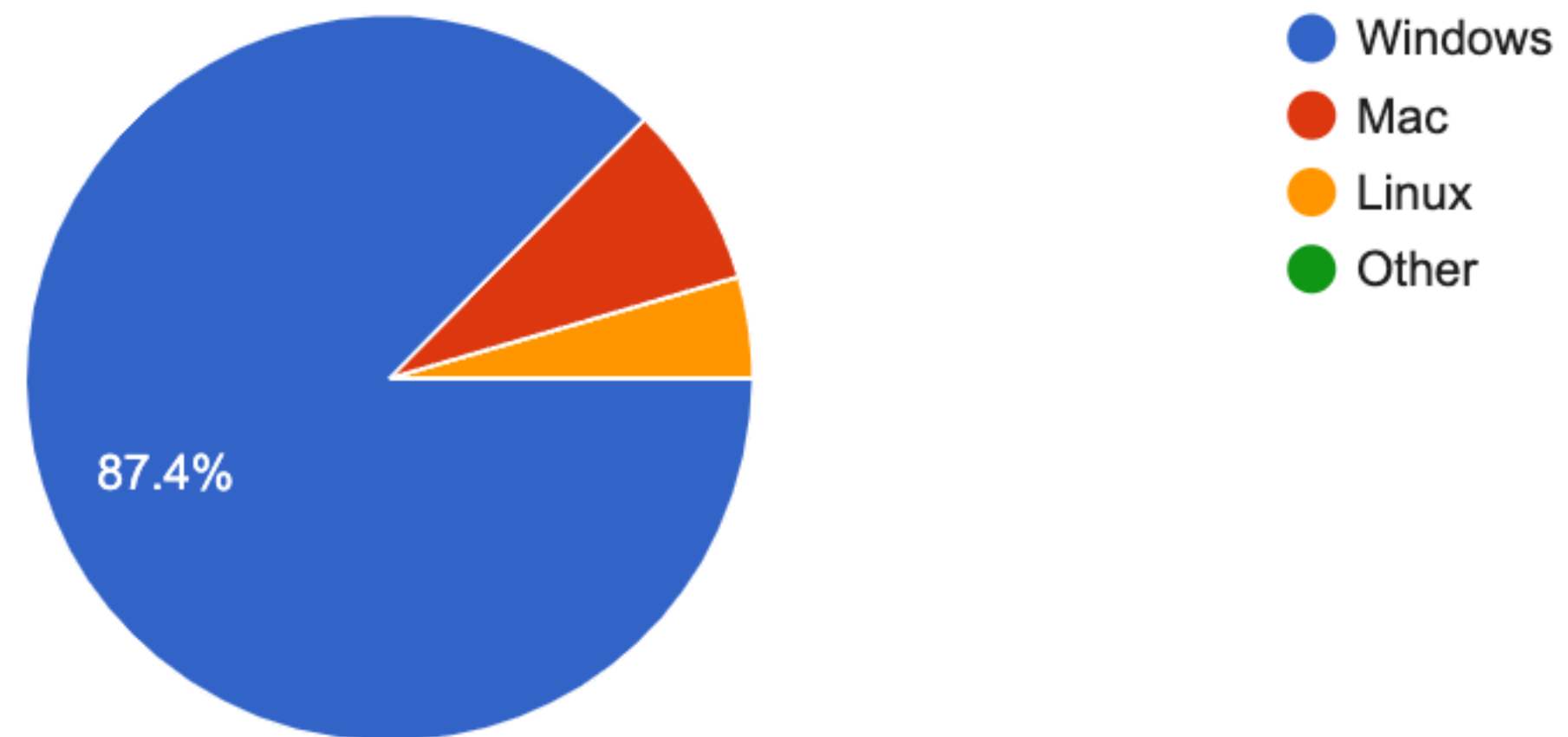
85 responses



Responses from 2020

What operating system is your development machine using?

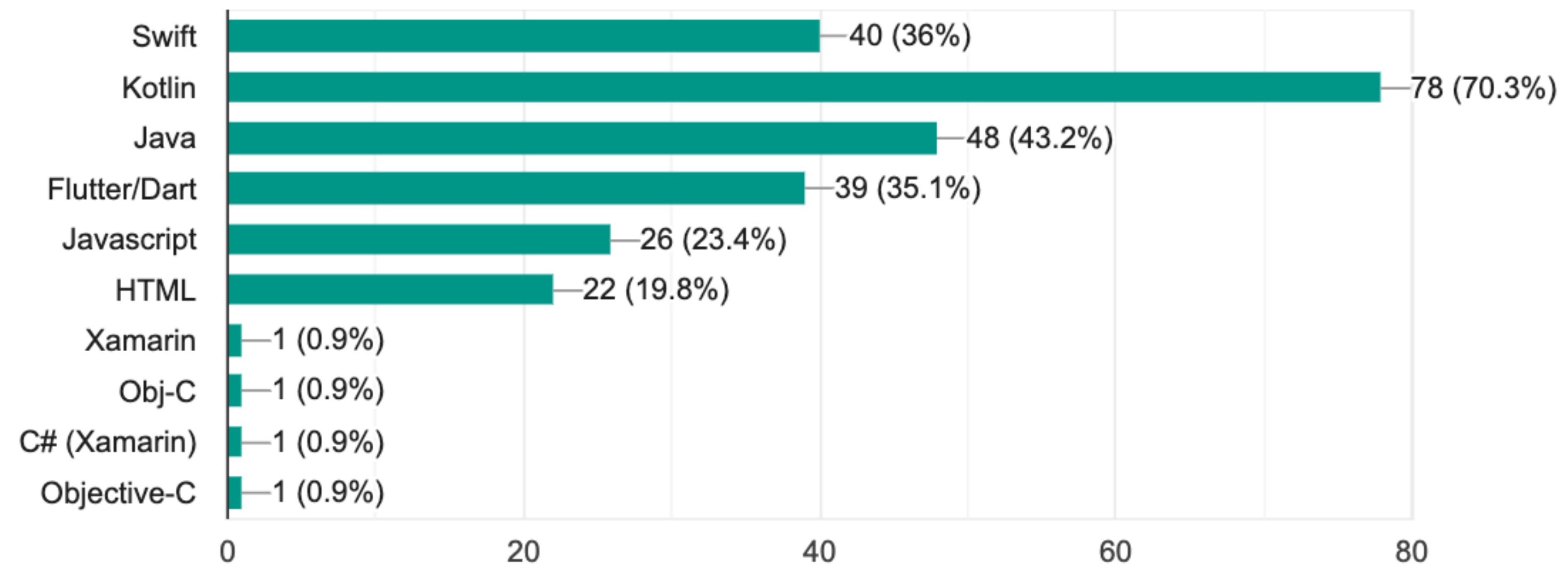
111 responses



Responses from 2020

What language would you like to use/learn?

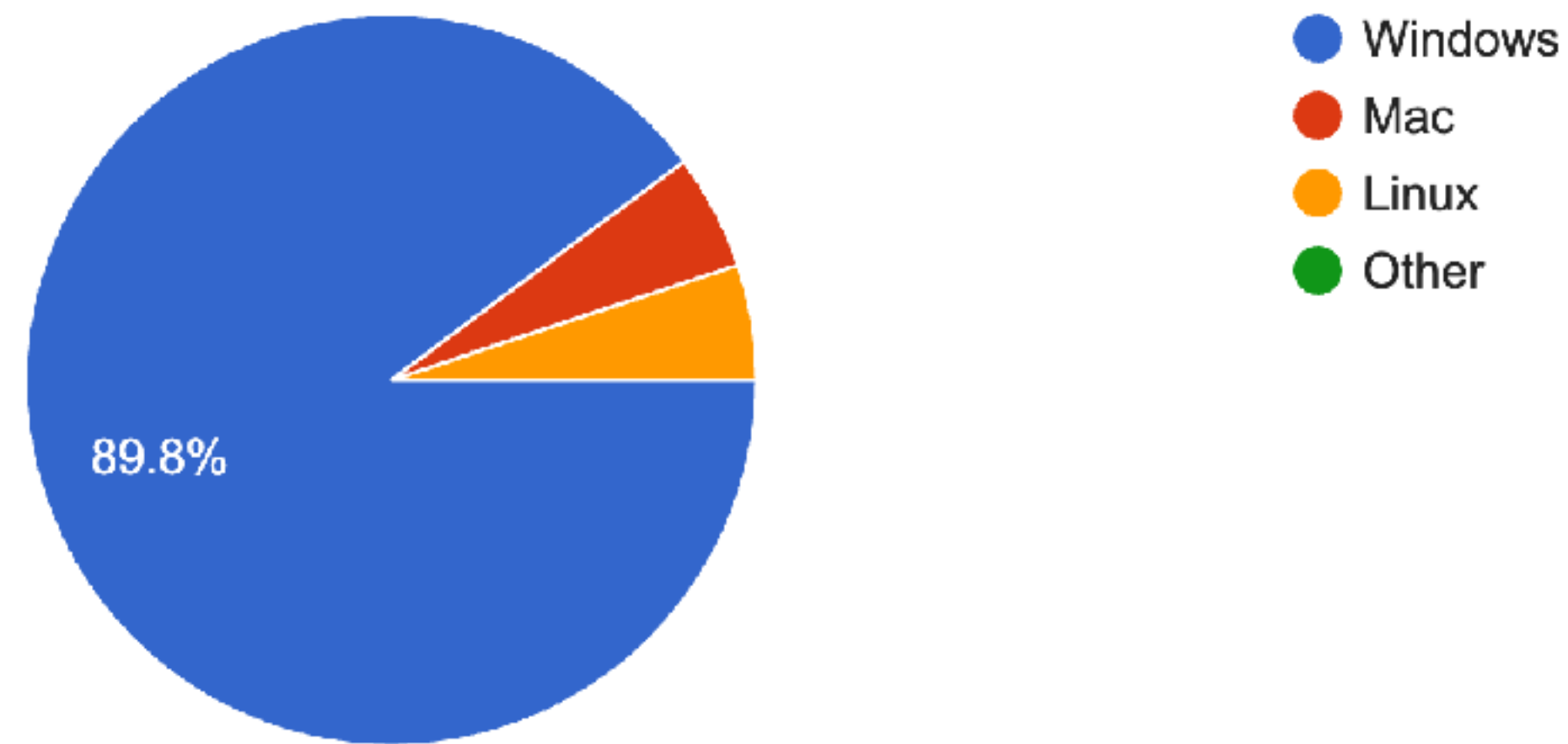
111 responses



Responses from 2021

What operating system is your development machine using?

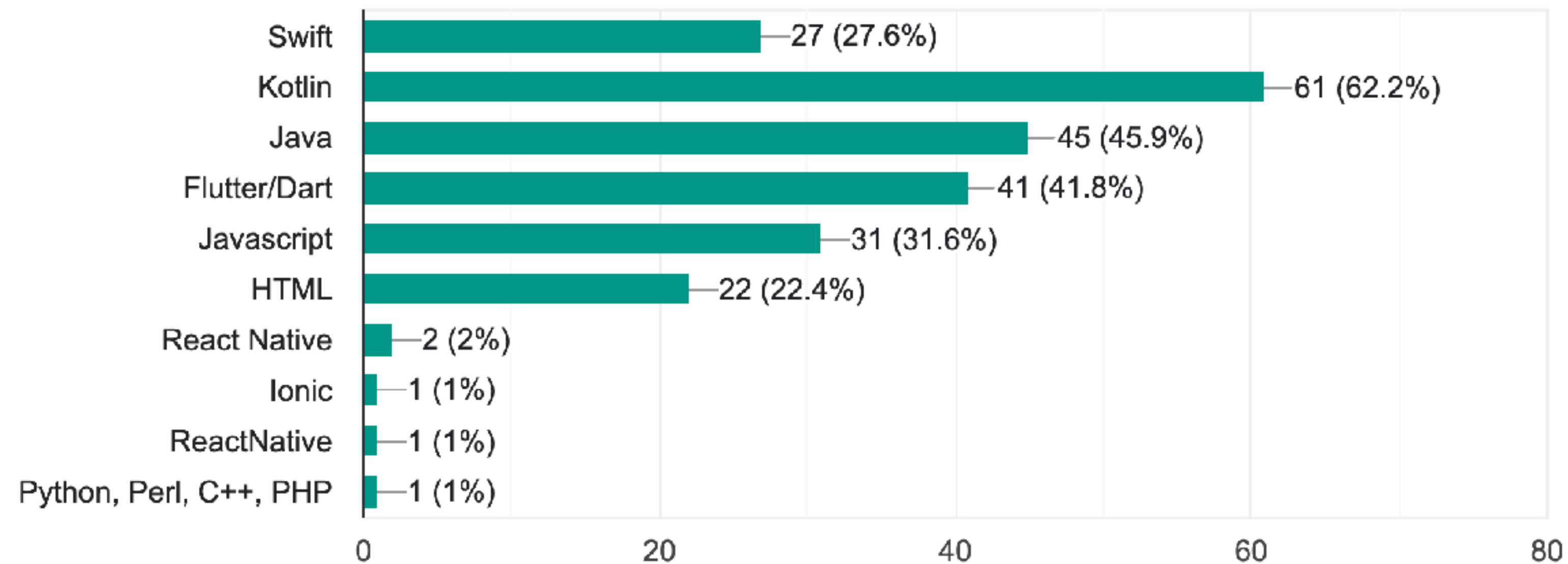
98 responses



Responses from 2021

What language would you like to use/learn?

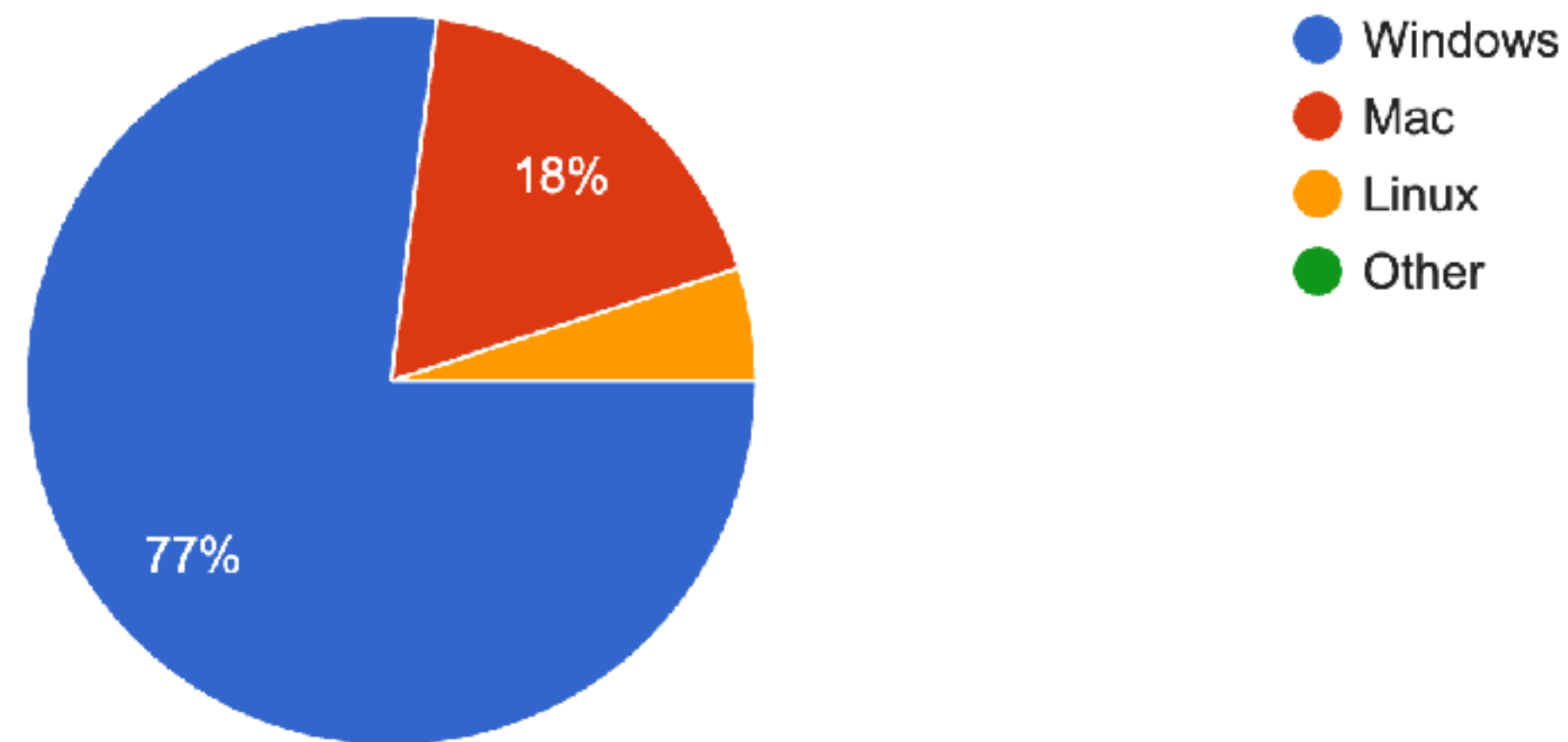
98 responses



Responses from 2022

What operating system is your development machine using?

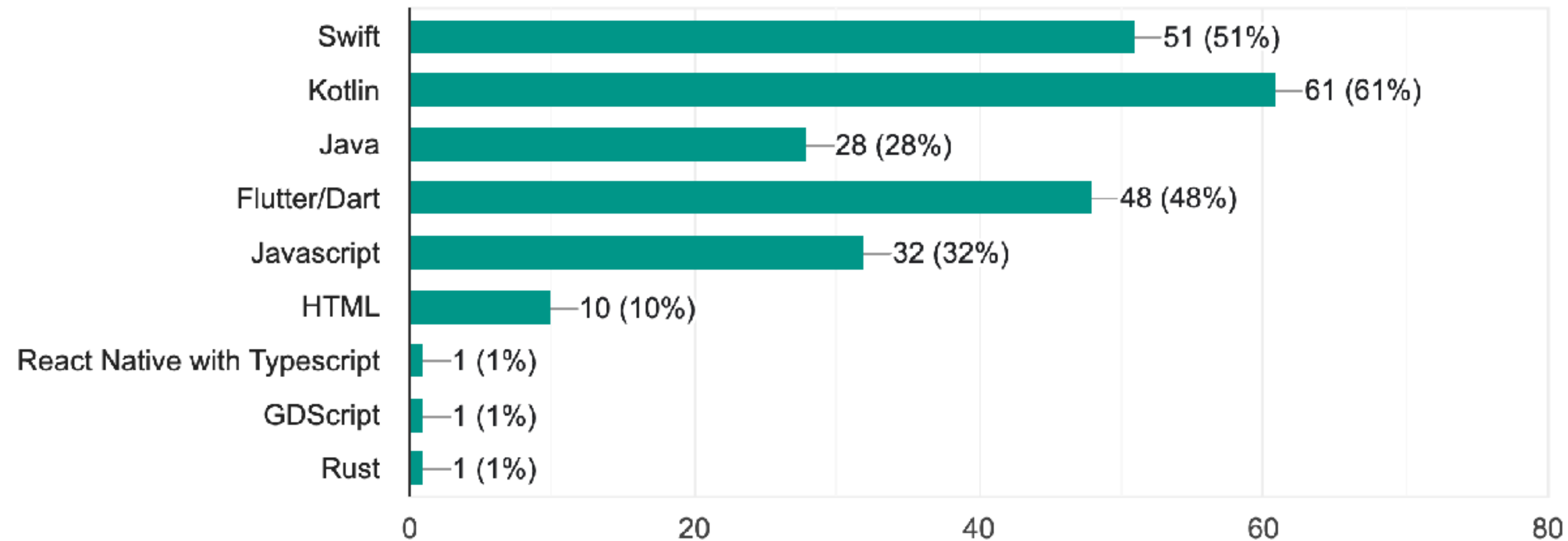
100 responses



Responses from 2022

What language would you like to use/learn?

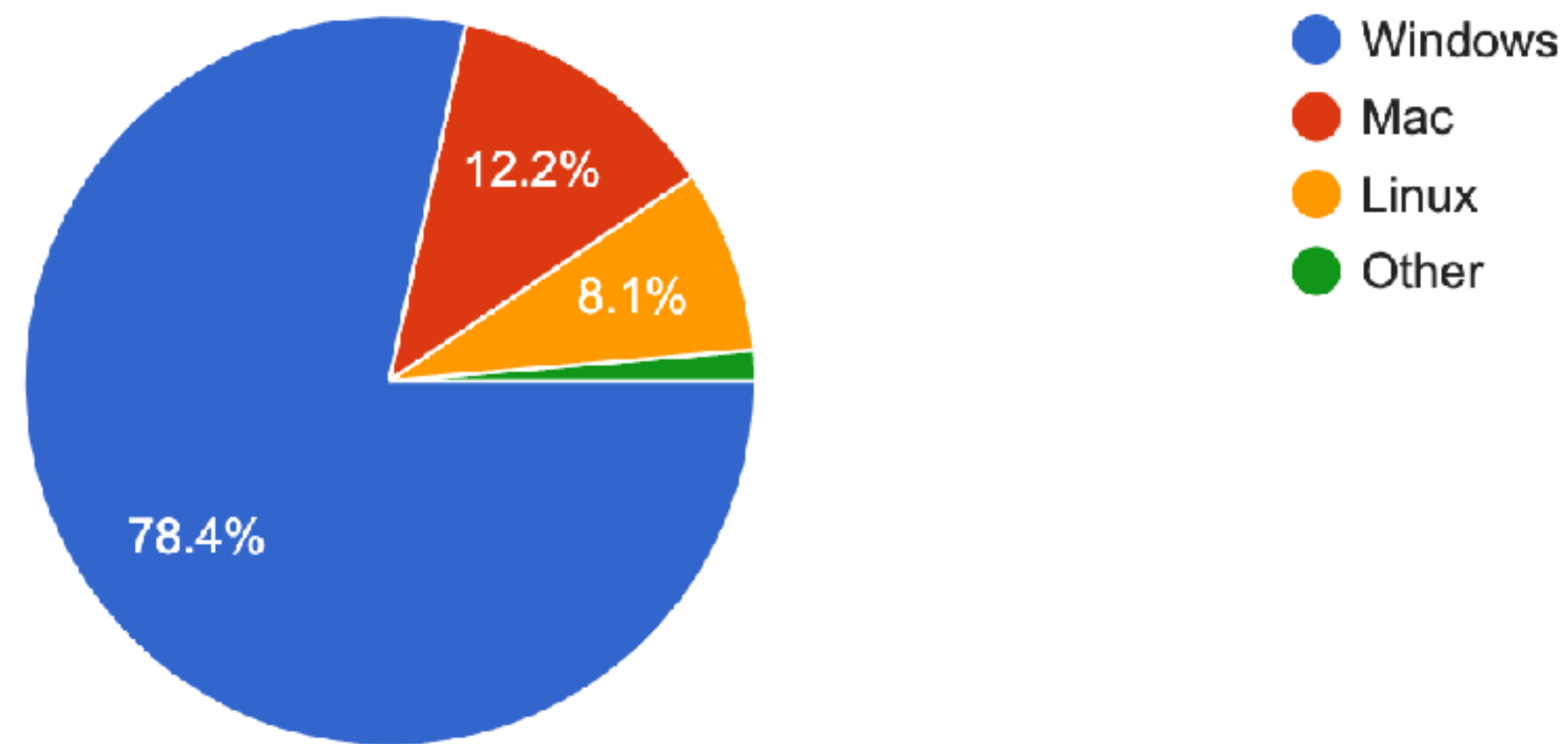
100 responses



Responses from 2023

What operating system is your development machine using?

74 responses

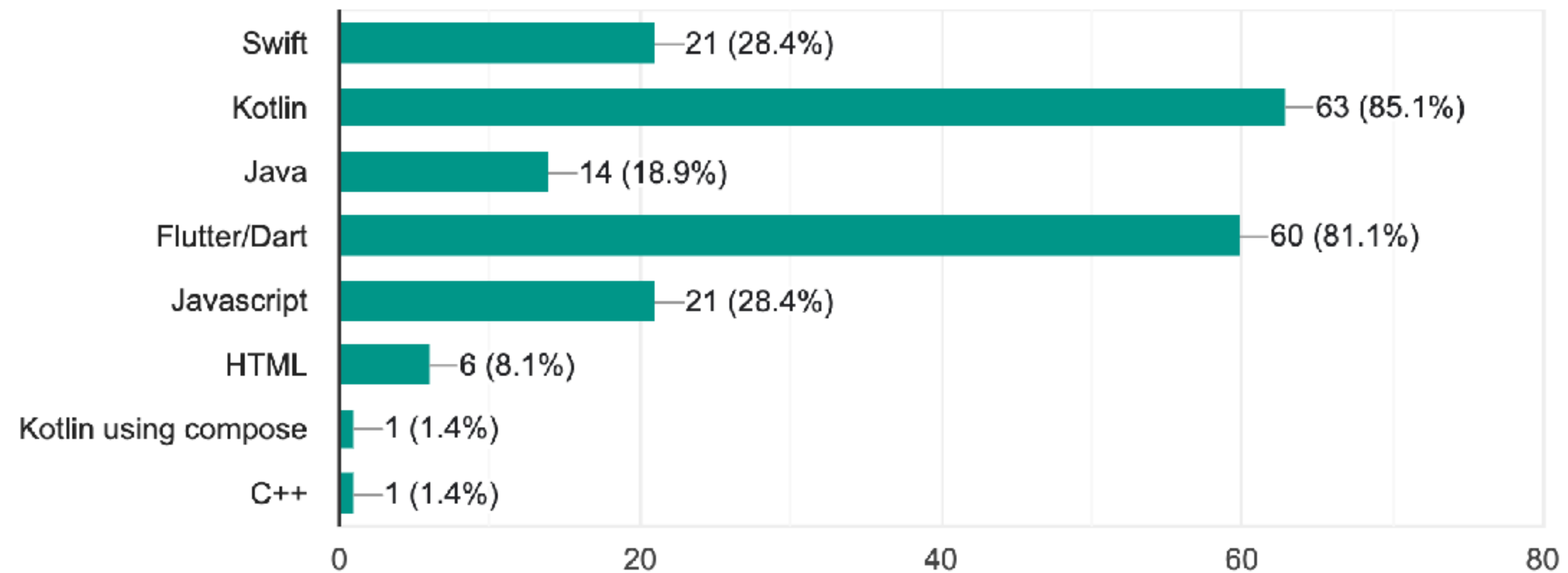


Responses from 2023

What language would you like to use/learn?



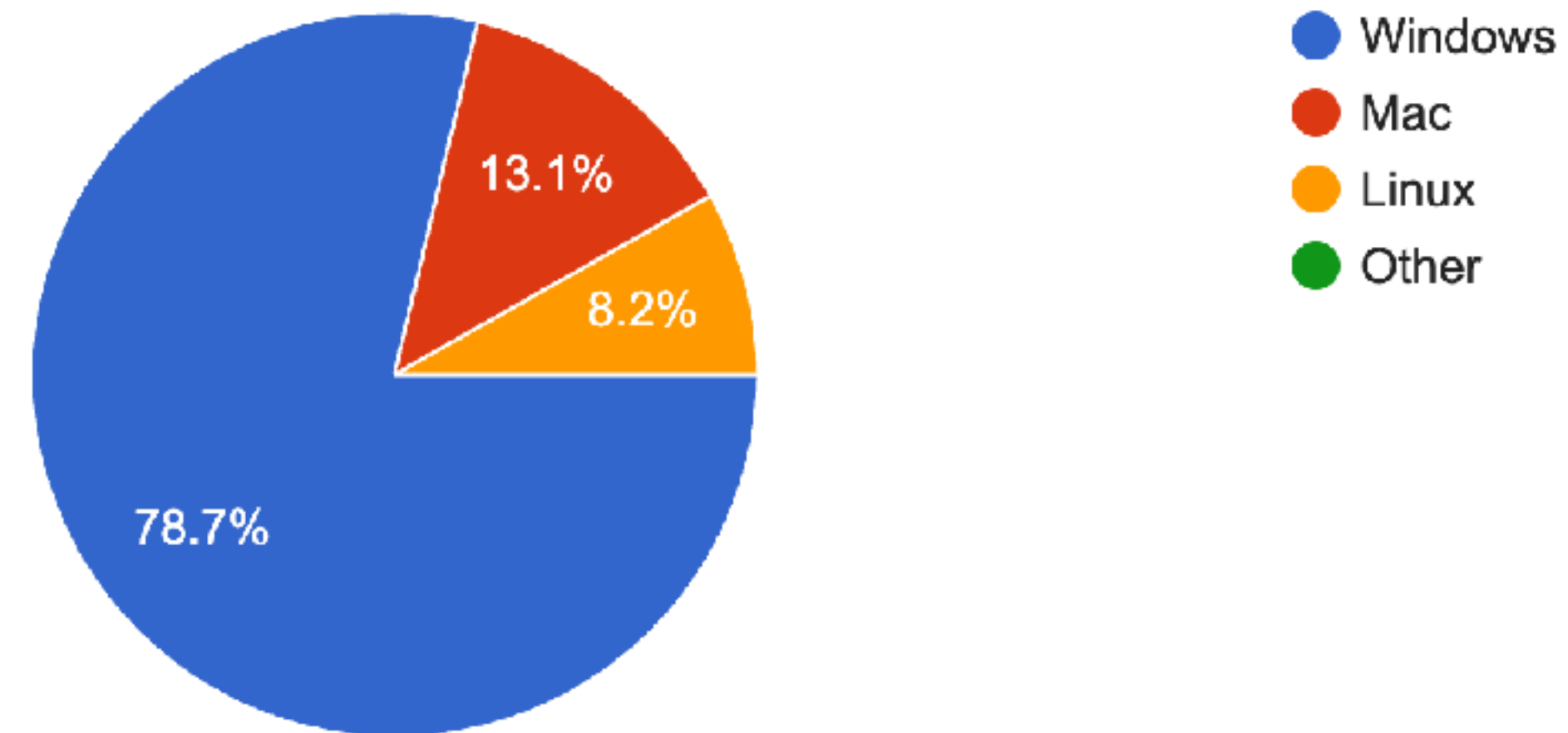
74 responses



Responses from 2024

What operating system is your development machine using?

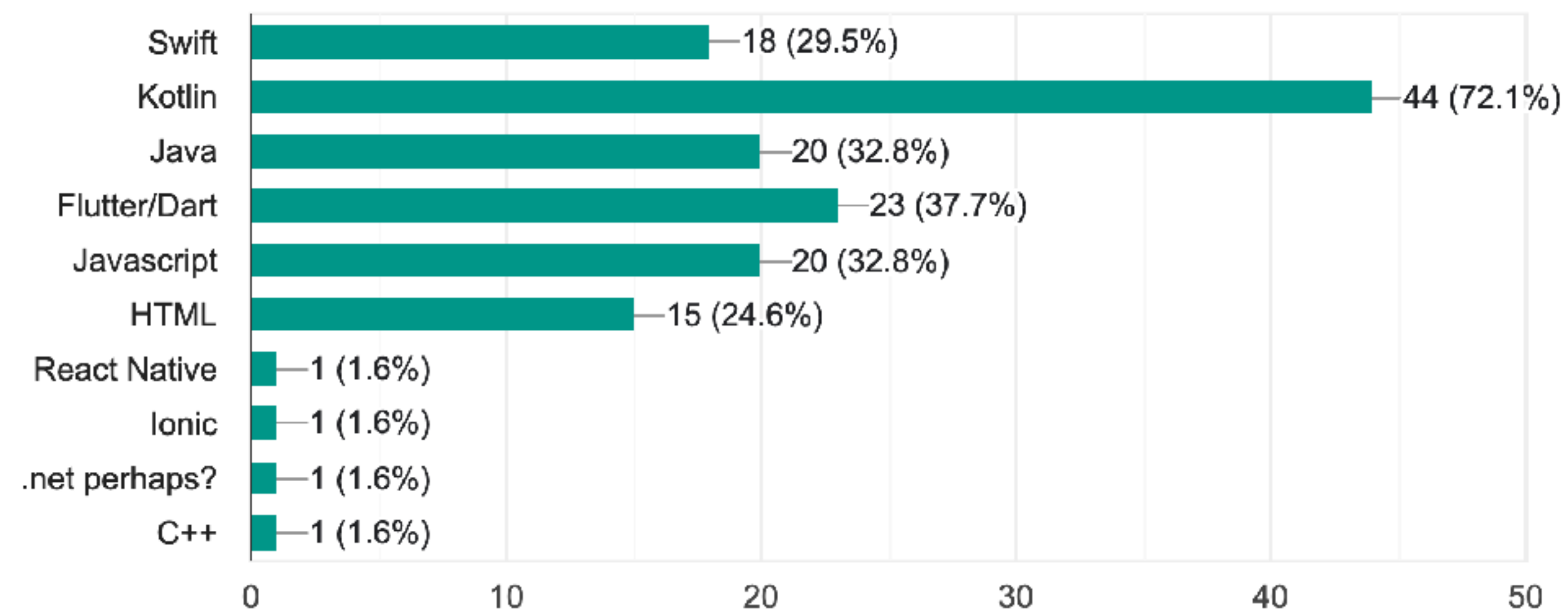
61 responses



Responses from 2023

What language would you like to use/learn?

61 responses



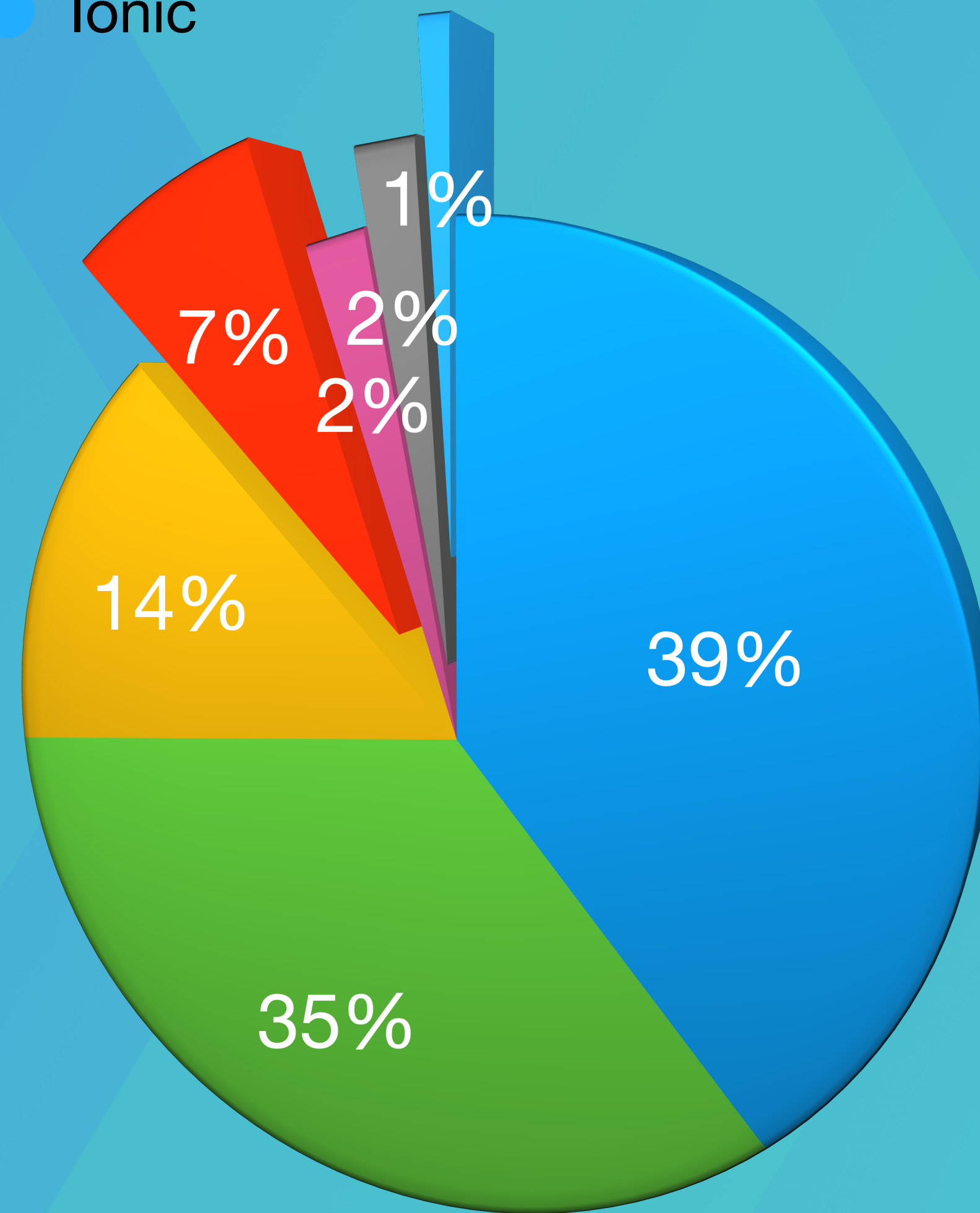
Check the quiz



Previous Year

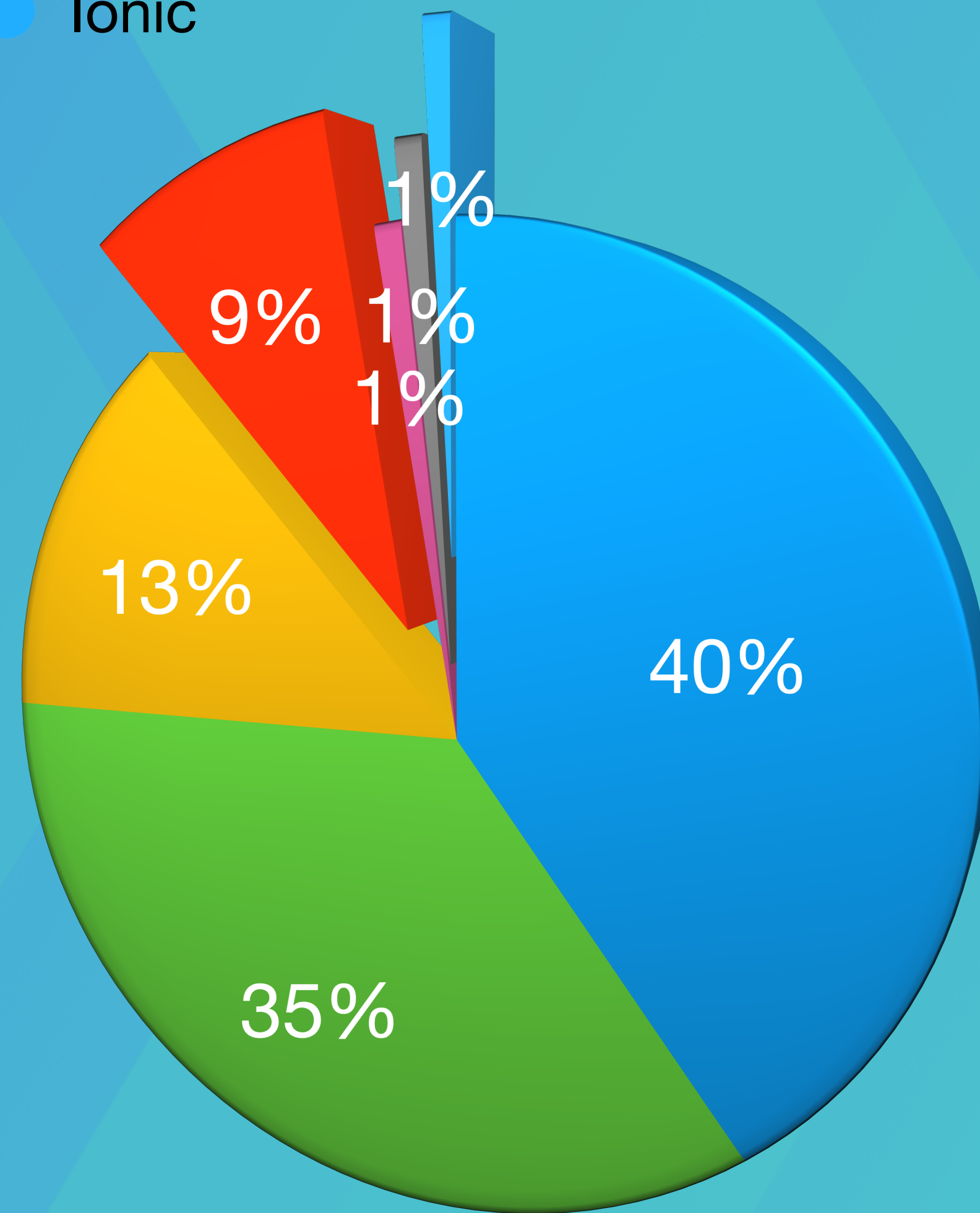


● Kotlin ● Flutter ● ReactNative ● Java ● Xamarin
● Swift ● Ionic



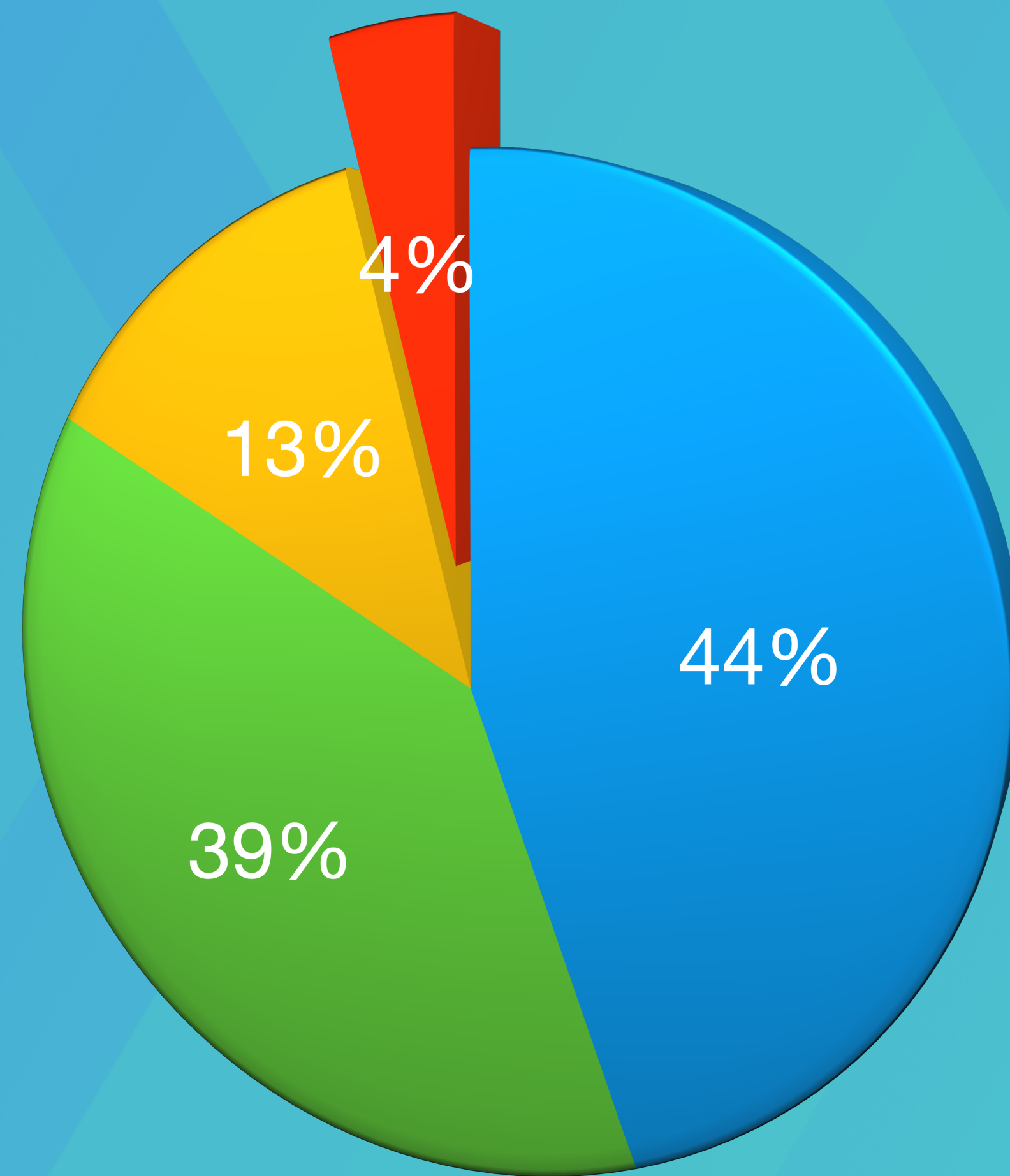
2019

● Kotlin ● Flutter ● ReactNative ● Java ● Xamarin
● Swift ● Ionic



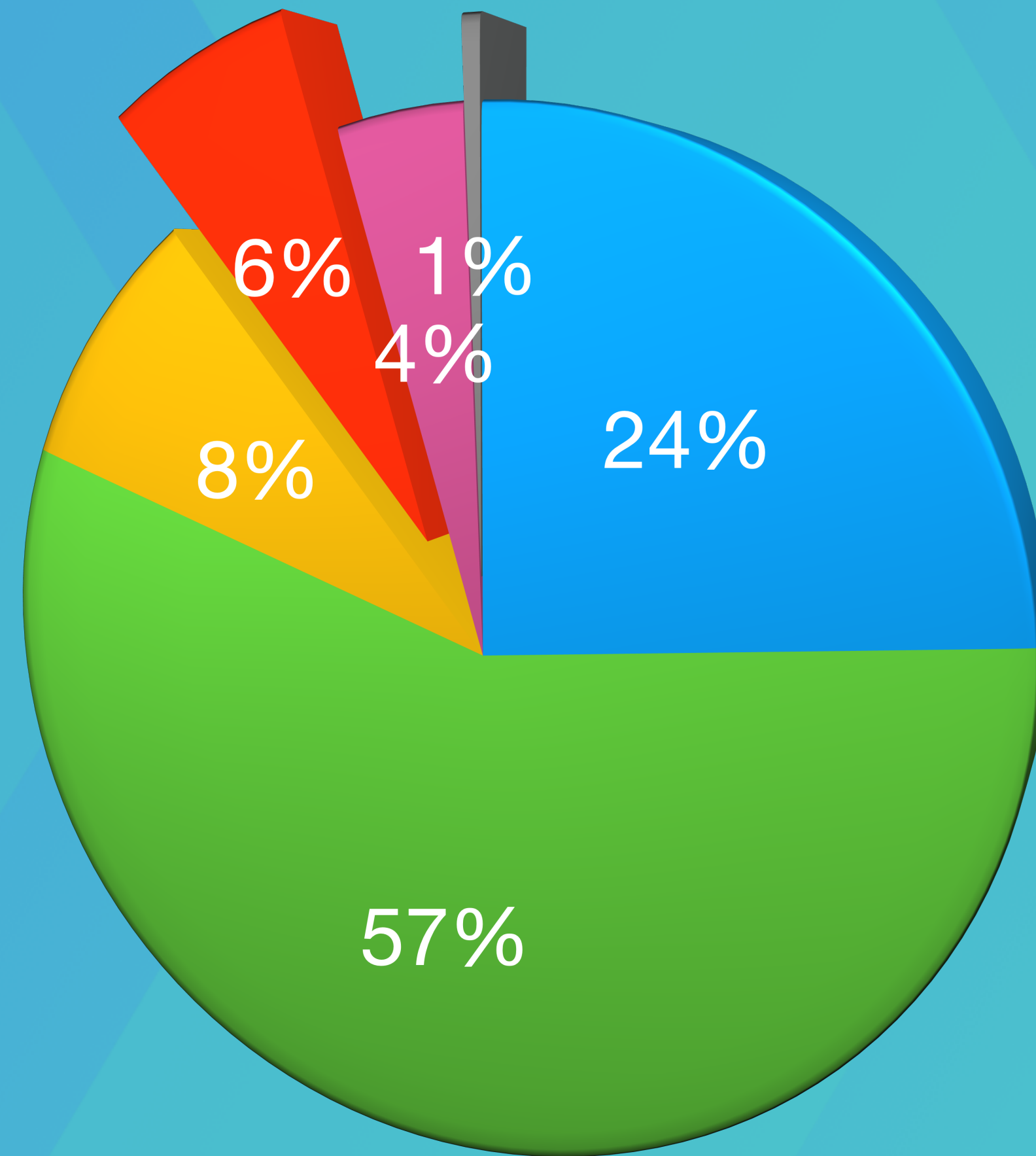
2020

● Kotlin ● Flutter ● ReactNative ● Java



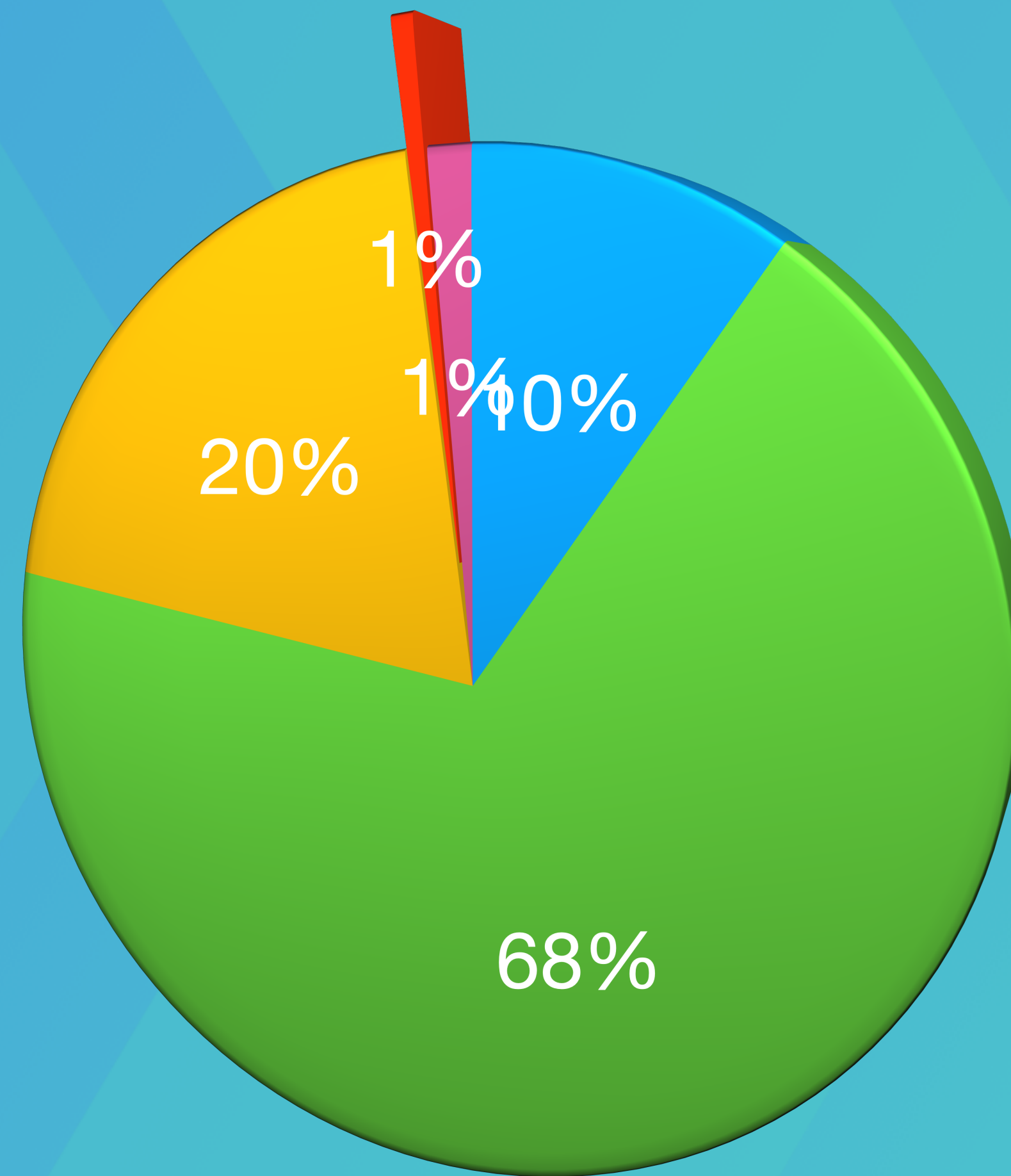
2021

● Kotlin ● Flutter ● ReactNative ● Java ● iOS
● Angular



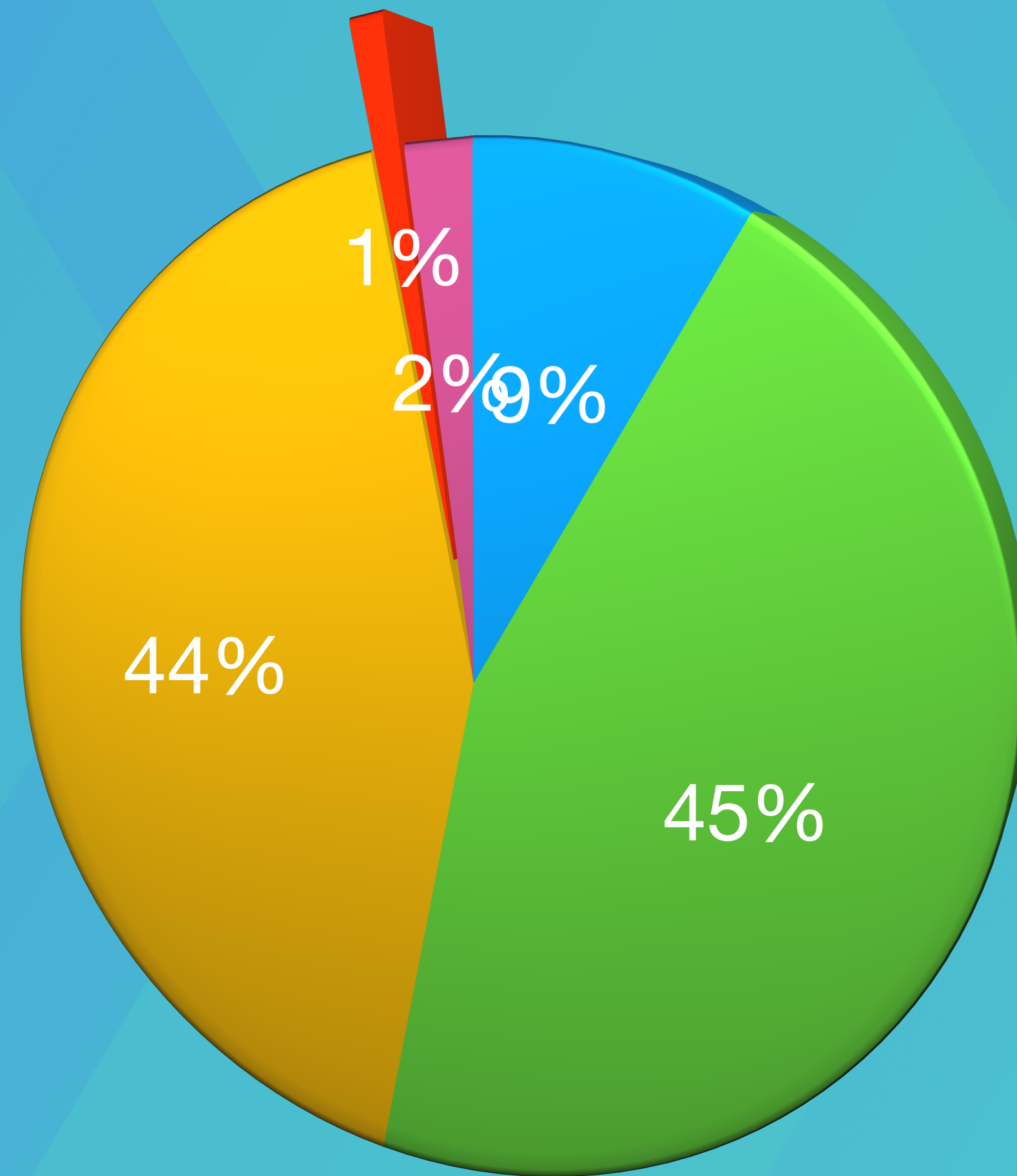
2022

● Kotlin ● Flutter ● ReactNative ● Java ● iOS

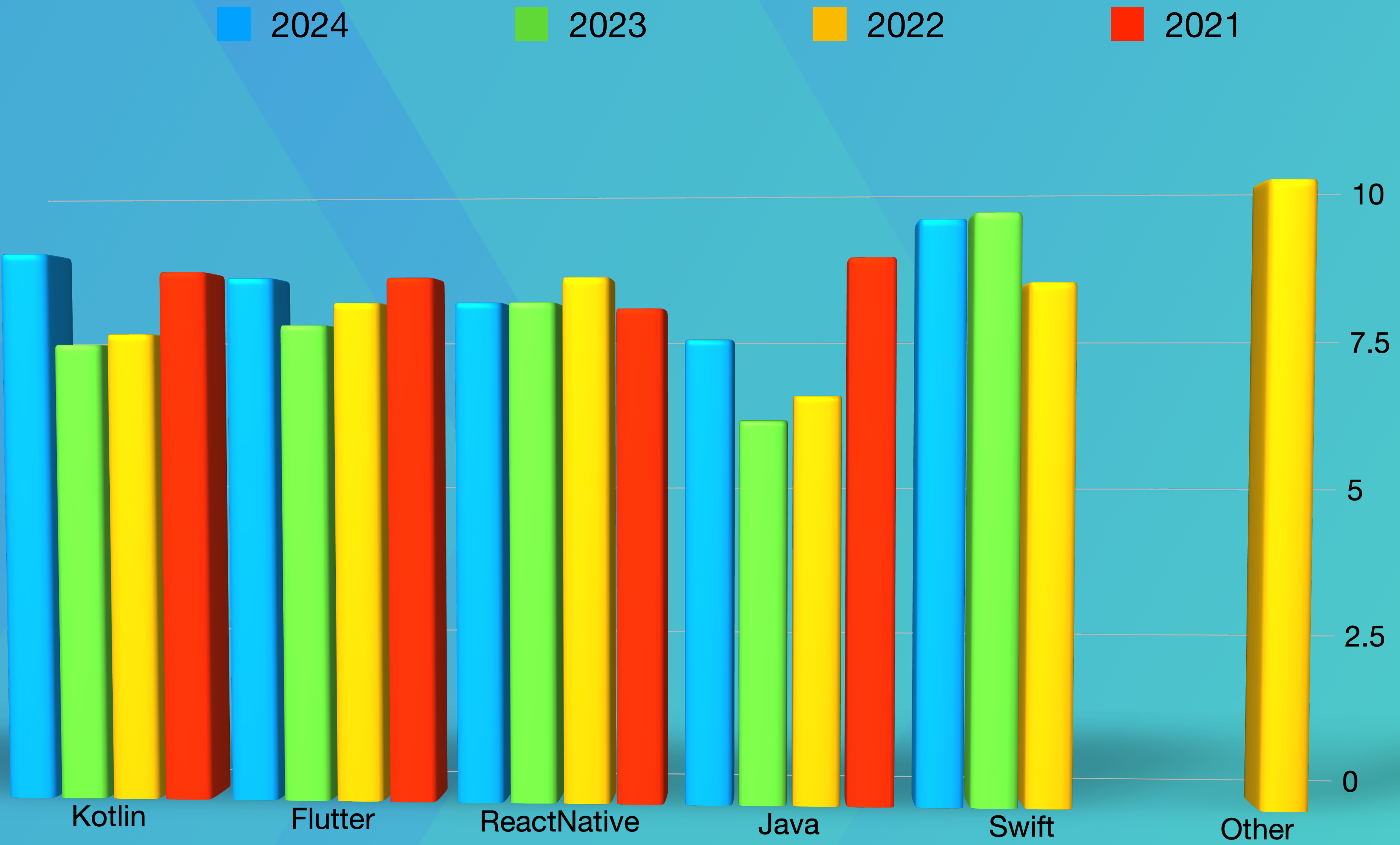


2023

● Kotlin ● Flutter ● ReactNative ● Java ● iOS



2024



This year



OBJ-C



Why Kotlin

- Modern programming language
- Object oriented
- Lambdas, Coroutines, Properties
- Since 2011
- Open Sources 2012
- Official First Class Android Citizen since 2017
- IntelliJ and Android Studio 3.0+



Why Kotlin



```
public class Aquarium {  
  
    private int mTemperature;  
  
    public Aquarium() { }  
  
    public int getTemperature() {  
        return mTemperature;  
    }  
  
    public void setTemperature(int mTemperature) {  
        this.mTemperature = mTemperature;  
    }  
  
    @Override  
    public String toString() {  
        return "Aquarium{" +  
            "mTemperature=" + mTemperature +  
            '}';  
    }  
}
```

Java

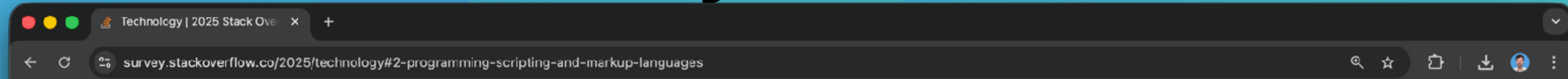
Why Kotlin

```
class Aquarium (var temperature: Int = 0)
```

Kotlin equivalent



Why Kotlin



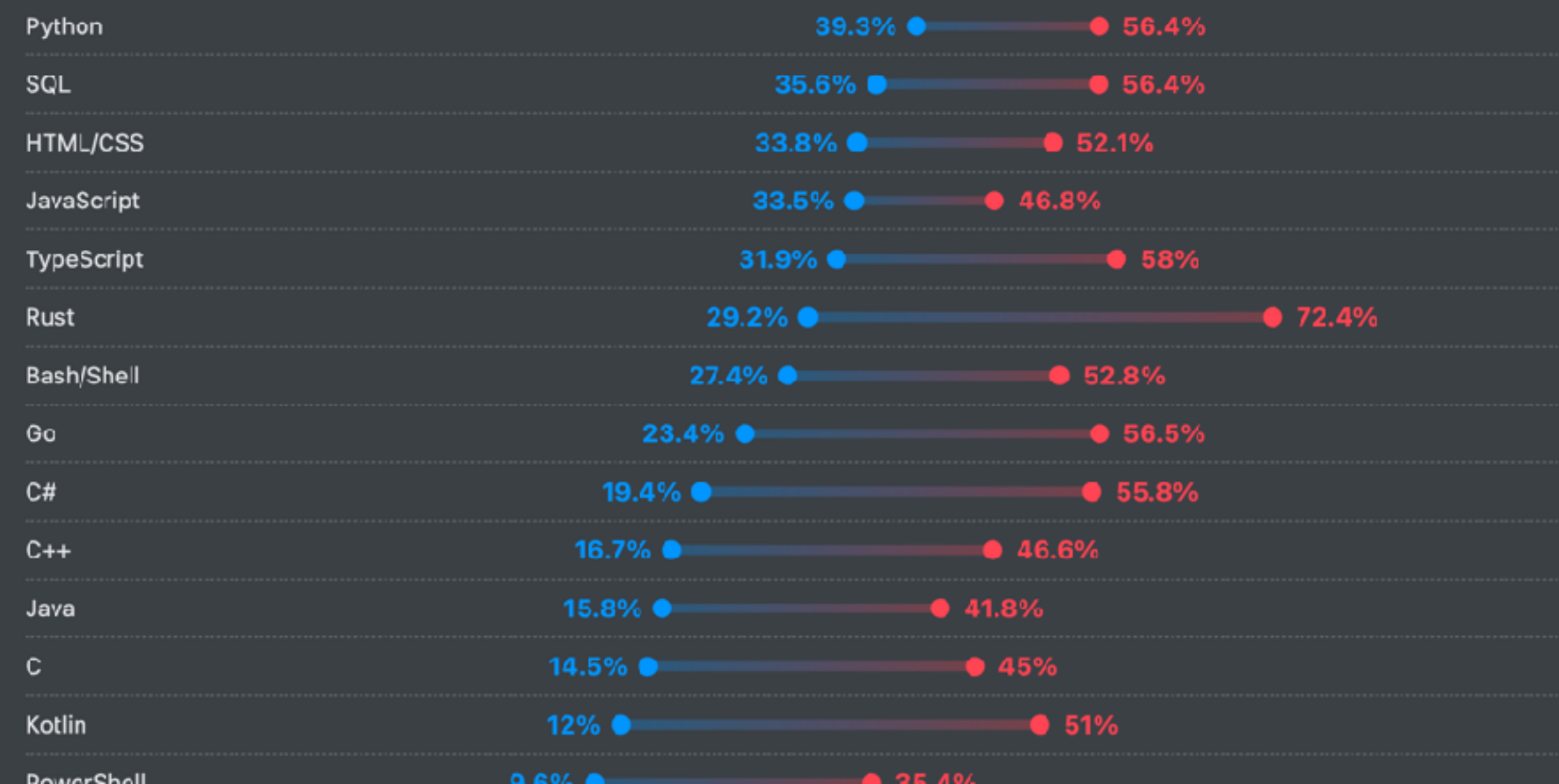
Admired and Desired 2.2

To better gauge hype versus reality, we created a visualization that shows the distance between the proportion of respondents who want to use a technology ("desired") and the proportion of users that have used the same technology in the past year and want to continue using it ("admired").

Programming, scripting, and markup languages

Rust is yet again the most admired programming language (72%), followed by Gleam (70%), Elixir (66%) and Zig (64%). Gleam is a new addition to the list, and for good reason - developers like it!

? Which **programming, scripting, and markup languages** have you done extensive development work in over the past year, and which do you want to work in over the next year? (If you both worked with the language and want to continue to do so, please check both boxes in that row.)



Why Kotlin

Kotlin v2.0.20

Solutions Docs Community Teach Play

Home

Get started

Take Kotlin tour

Kotlin overview

Kotlin Multiplatform

Kotlin for server side

Kotlin for Android

Kotlin Wasm

Kotlin Native

Kotlin for JavaScript

Kotlin for data analysis

Kotlin for competitive programming

What's new in Kotlin

Kotlin evolution and roadmap

Basics

Concepts

Multiplatform development

Data analysis

Platforms

Standard library

Official libraries

API reference

Language reference

Tools

Compiler and plugins

Learning materials

Early access preview (EAP)

Other resources

Kotlin overview / Kotlin Multiplatform

Kotlin Multiplatform

Edit page Last modified: 25 September 2024

The Kotlin Multiplatform technology is designed to simplify the development of cross-platform projects. It reduces time spent writing and maintaining the same code for different platforms while retaining the flexibility and benefits of native programming.

Server

Android

iOS

Desktop

Web

Client Multiplatform

Common Multiplatform

Kotlin Multiplatform use cases

Android and iOS applications

Sharing code between mobile platforms is a major [Kotlin Multiplatform](#) use case. With Kotlin Multiplatform, you can build cross-platform mobile applications that share code between Android and iOS projects to implement networking, data storage and data validation, analytics, computations, and other application logic.

Check out the [Get started with Kotlin Multiplatform](#) and [Create a multiplatform app using Ktor and SQLDelight](#) tutorials, where you will create applications for Android and iOS that include a module with shared code for both platforms.

Thanks to [Compose Multiplatform](#), a Kotlin-based declarative UI framework developed by JetBrains, you can also share UIs across Android and iOS to create fully cross-platform apps:

UI

Kotlin Multiplatform

Kotlin Multiplatform use cases

Android and iOS applications

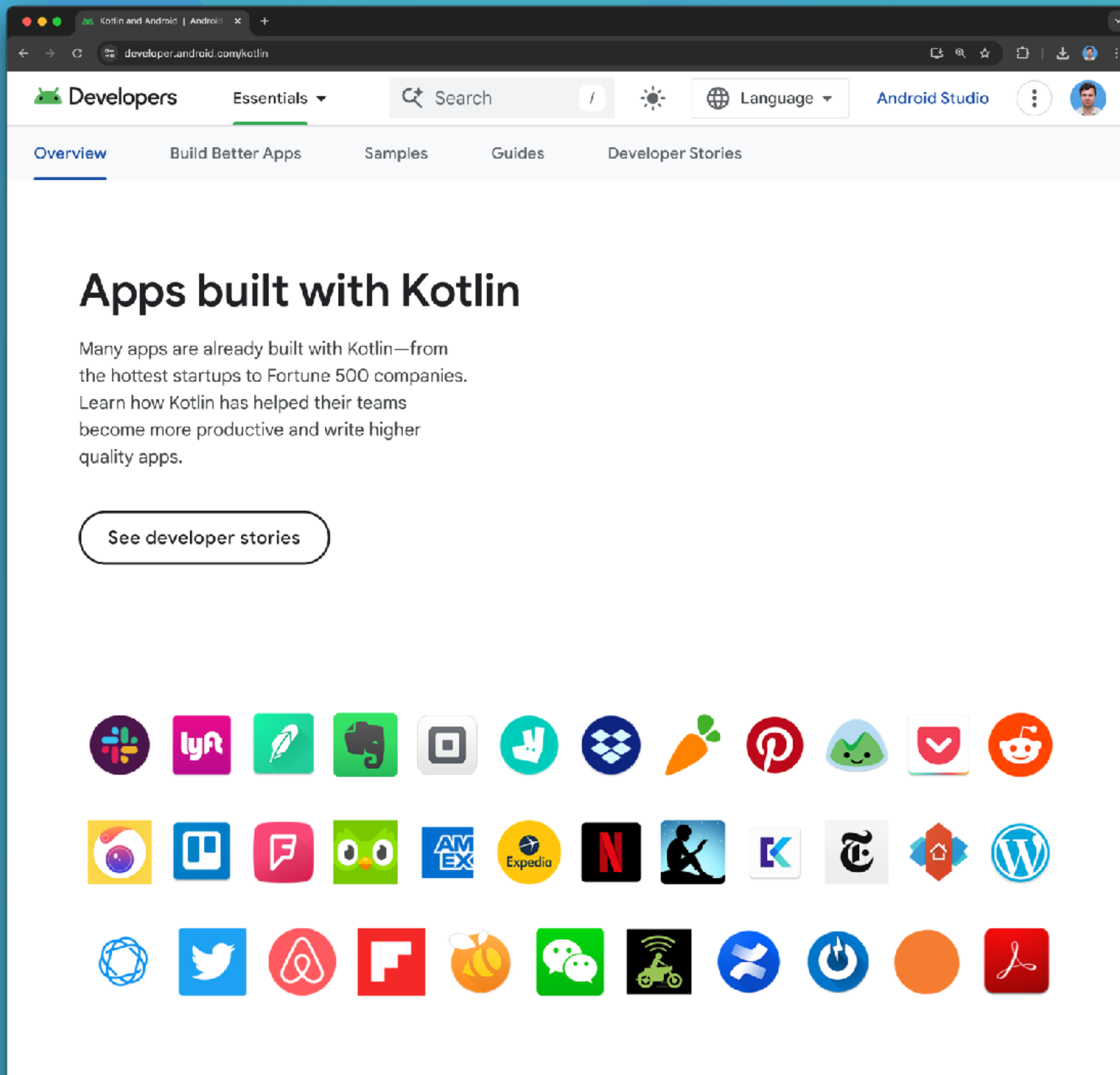
Multiplatform libraries

Desktop applications

Code sharing between platforms

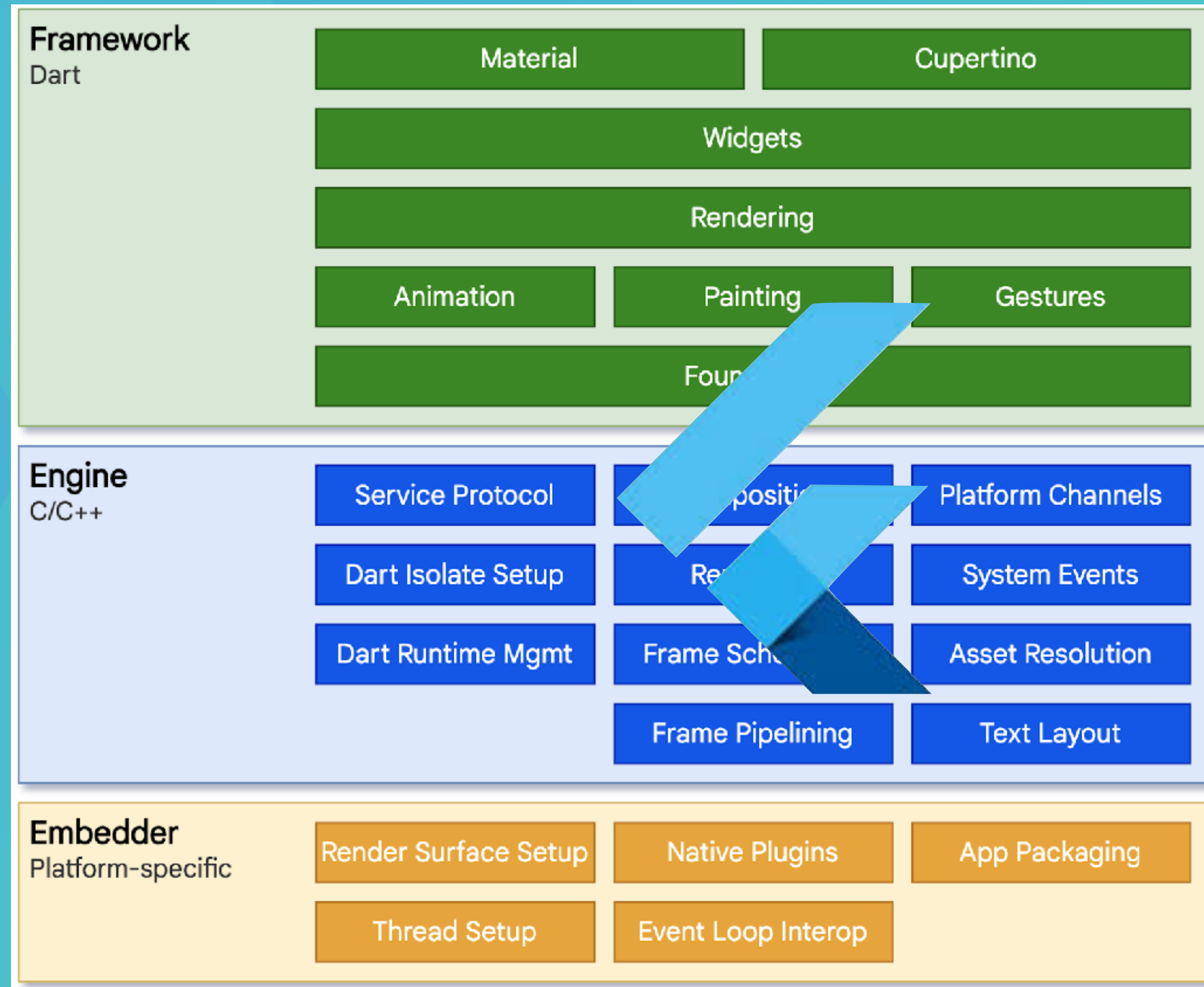
Get started

Sample projects



Why Flutter

- Platform-agnostic
- Simplifies and speeds application development
- Easy to learn and easy to use
- Scales well
- Offer an excellent user experience



Why Flutter

Flutter
Software

React Native
Software

+ Add comparison

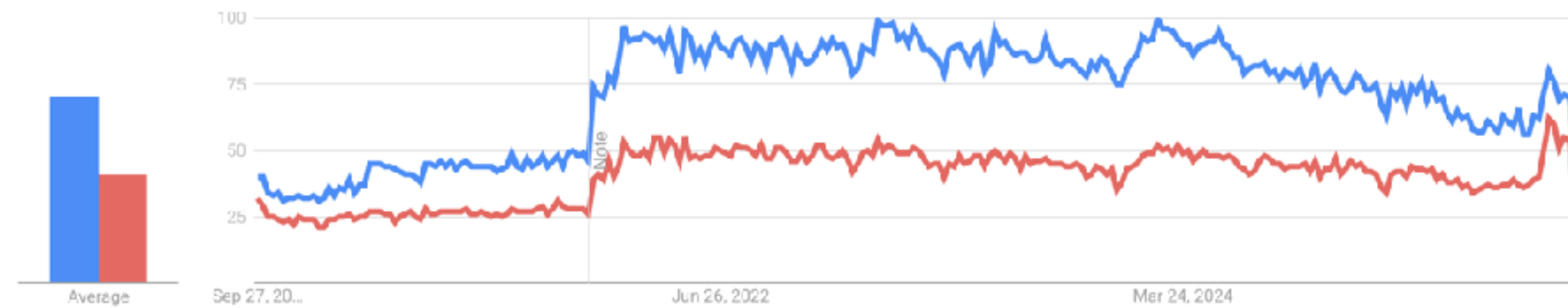
Worldwide ▾

Past 5 years ▾

All categories ▾

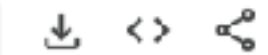
Web Search ▾

Interest over time ?



Compared breakdown by region

Region ▾



Flutter React Native

Sort: Interest for Flutter ▾

1 Yemen



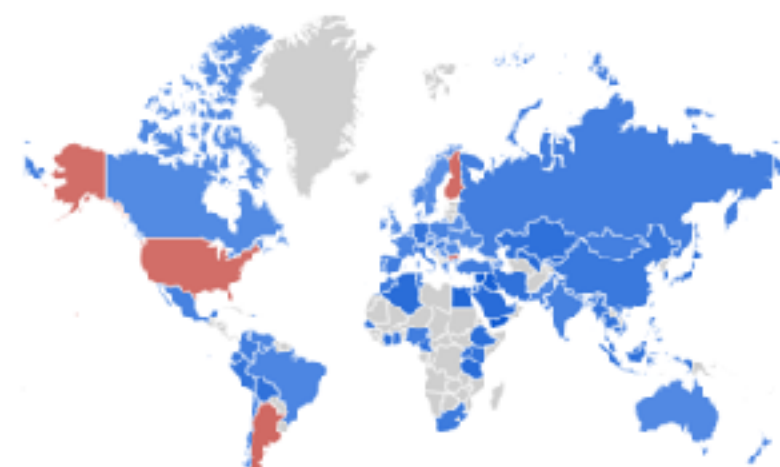
2 Syria



3 Iraq



4 Egypt



Flutter
Software

React Native
Software

+ Add comparison

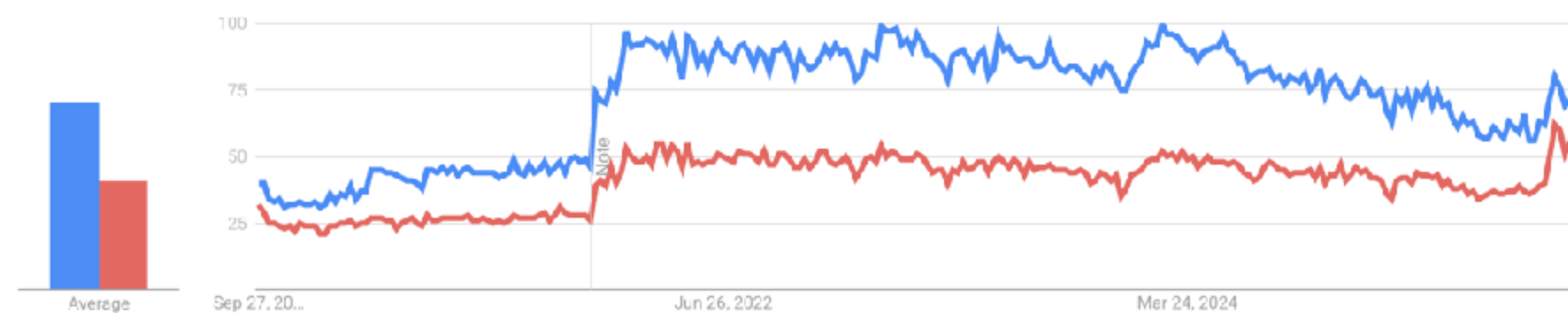
Worldwide ▼

Past 5 years ▼

All categories ▼

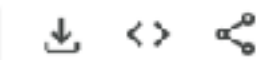
Web Search ▼

Interest over time ?



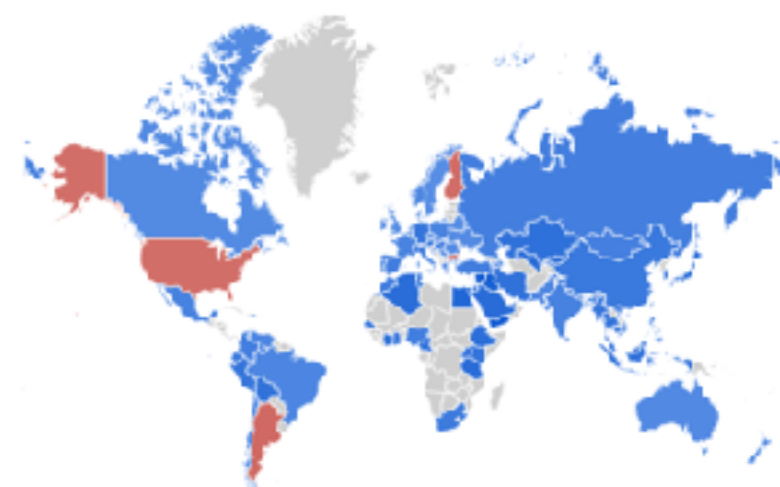
Compared breakdown by region

Region ▼



Flutter React Native

Sort: Interest for Flutter ▼



Color intensity represents percentage of searches [LEARN MORE](#)

< Showing 1-5 of 90 regions >

☐ Include low search volume regions

1 Yemen



2 Syria



3 Iraq



4 Egypt



5 Japan



Flutter
Software

React Native
Software

+ Add comparison

Romania ▼

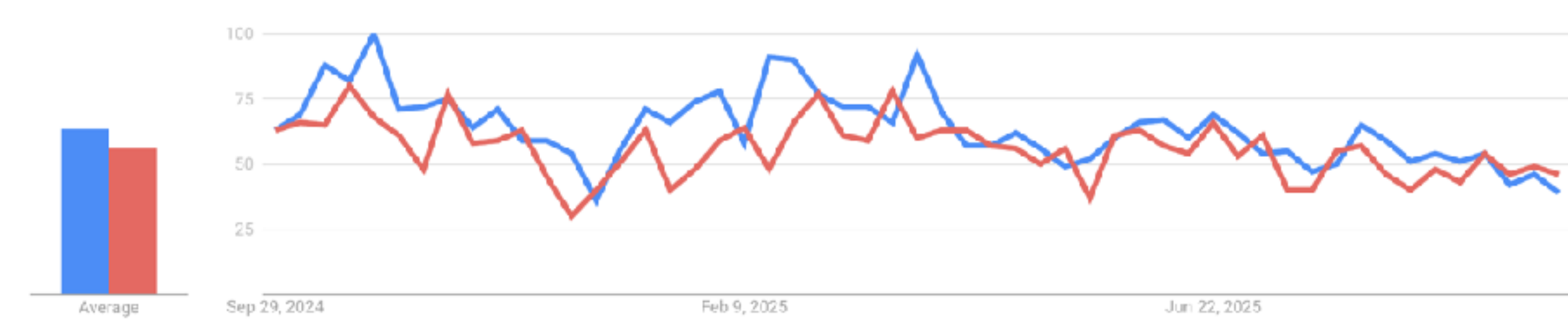
Past 12 months ▼

All categories ▼

Web Search ▼

Interest over time ?

Download Code Share

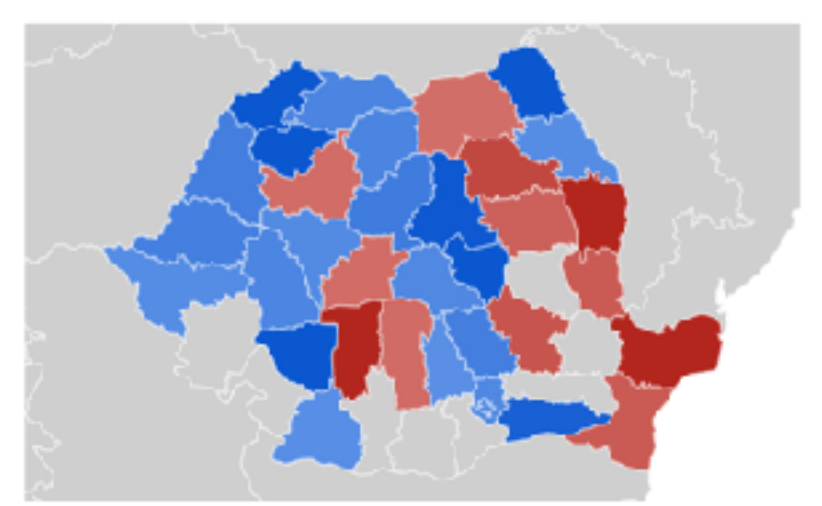


Compared breakdown by subregion

Subregion ▼ Download Code Share

Flutter React Native

Sort: Interest for Flutter ▼



1	Covasna County	
2	Satu Mare County	
3	Sălaj	
4	Botoșani County	
5	Gorj County	

Color intensity represents percentage of searches [LEARN MORE](#)

< Showing 1-5 of 34 subregions >

Why Flutter

Repository search results

github.com/search?q=react-native&type=repositories

Filter by

<> Code

Repositories

Issues

Pull requests

Discussions

Users

More

10M

561k

451k

1M

15k

436

Languages

JavaScript

TypeScript

Python

Java

Objective-C

Kotlin

HTML

Swift

C++

561k results (558 ms)

Sort by: Best match

Save

...

React Native

React Native is a JavaScript mobile framework developed by Facebook.

github.com

Wikipedia

facebook/react-native

View topic

facebook/react-native

A framework for building native applications using React

react

android

ios

mobile

react-native

C++

124k

Updated 48 minutes ago

react-native-elements/react-native-elements

Cross-Platform React Native UI Toolkit

react

android

ios

ui

react-native

Star

Sponsor

Repository search results

github.com/search?q=flutter&type=repositories

Filter by

<> Code

Repositories

Issues

Pull requests

Discussions

Users

More

23.9M

901k

424k

483k

12k

57k

Languages

Dart

C++

JavaScript

HTML

Makefile

Java

Objective-C

Kotlin

901k results (540 ms)

Sort by: Best match

Save

...

flutter/flutter

Flutter makes it easy and fast to build beautiful apps for mobile and beyond

android

windows

macos

dart

ios

Dart

173k

Updated 20 minutes ago

Unstar

flutter/plugins

Plugins for Flutter maintained by the Flutter team

android

plugin

dart

ios

flutter

Dart

17.6k

Updated on Feb 22, 2023

Star

iampawan/FlutterExampleApps

[Example APPS] Basic Flutter apps, for flutter devs.

android

dart

ios

material-design

cross-platform

Dart

21.3k

Updated on Aug 9, 2024

Star

Sponsor

Course Goals

- Knowledge of key base concepts for developing mobile applications.
- Learn the Android platform.
- Learn a framework to develop multi-platform applications (Android&iOS)



Lecture outcomes

- Understand the generated artifacts
- Lifecycle of applications, activities and fragments.
- Use logs to debug and study the behavior.

